Engineering Mathematics – III

Unit III : Complex Differentiation

Analytic Functions, Conformal Mappings, Bilinear Transformations and Applications

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1 Complex Differentiation

Analytic Functions, Cauchy–Riemann Equations and Harmonic Functions

1.1 Complex Functions and Differentiability

Class of 24/08/26

We know that $z = x + iy$ is a complex variable and we can construct a complex function $w = f(z)$ using the complex variable.

Example

$z = x + iy$ is a complex variable.

$$f(z) = z^2 = (x+iy)^2 = (x^2 - y^2) + i\,2xy$$
$$w = f(z) = u(x,y) + i\,v(x,y)$$

Like the real function $y = f(x)$, we can define the *limit* and *continuity* of $w = f(z)$.

Differentiability

The derivative of the function $f(z)$ is defined by

$$f'(z) = \lim_{\Delta z \to 0} \frac{f(z+\Delta z) - f(z)}{\Delta z},$$

provided the limit exists and is the **same along all the paths** along which $\Delta z \to 0$.

Example 1: $f(z) = z^2$ is differentiable

$$\begin{aligned} f(z+\Delta z) &= (z+\Delta z)^2 \\ \implies f(z+\Delta z) - f(z) &= (z+\Delta z)^2 - z^2 \\ &= \cancel{z^2} + (\Delta z)^2 + 2z\,\Delta z - \cancel{z^2} \\ &= (\Delta z)^2 + 2z\,\Delta z \end{aligned}$$

$$\boxed{f(z+\Delta z) - f(z) = \Delta z\,(\Delta z + 2z)}$$

Now

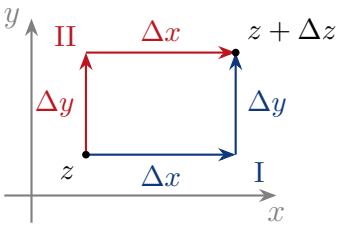
$$f'(z) = \lim_{\Delta z \to 0} \frac{f(z+\Delta z) - f(z)}{\Delta z} = \lim_{\Delta z \to 0} \frac{\cancel{\Delta z}\,(\Delta z + 2z)}{\cancel{\Delta z}} = \lim_{\Delta z \to 0} (\Delta z + 2z)$$
$$f'(z) = \lim_{(\Delta x + i\Delta y) \to 0} \big[(\Delta x + i\Delta y) + 2z\big] \qquad (1)$$

Along path I: $\Delta y \to 0$ first and then $\Delta x \to 0$.

$$(1) \Rightarrow f'(z) = \lim_{\Delta x \to 0} (\Delta x + 2z) \;\Rightarrow\; \boxed{f'(z) = 2z} \quad — (2)$$

Along path II: $\Delta x \to 0$ first and then $\Delta y \to 0$.

$$(1) \Rightarrow f'(z) = \lim_{\Delta y \to 0} (i\,\Delta y + 2z) \;\Rightarrow\; \boxed{f'(z) = 2z} \quad — (3)$$



By (2) and (3), $f'(z)$ is the same along the two paths where $\Delta z \to 0$.

$\therefore\ f(z) = z^2$ is a differentiable function and $f'(z) = 2z$.

Example 2: $f(z) = \bar{z}$ is not differentiable

$$\begin{aligned} f(z+\Delta z) &= \overline{z+\Delta z} \\ f(z+\Delta z) - f(z) &= \overline{(z+\Delta z)} - \bar{z} \\ &= \overline{(x+iy+\Delta x+i\Delta y)} - \overline{(x+iy)} \\ &= \overline{\big((x+\Delta x)+i(y+\Delta y)\big)} - (x-iy) \\ &= \big(x+\Delta x - i(y+\Delta y)\big) - (x - iy) \\ f(z+\Delta z) - f(z) &= \Delta x - i\,\Delta y \end{aligned}$$

Then

$$f'(z) = \lim_{\Delta z \to 0} \frac{f(z+\Delta z) - f(z)}{\Delta z} = \lim_{(\Delta x + i\Delta y)\to 0} \frac{\Delta x - i\Delta y}{\Delta x + i\Delta y} \qquad \text{— (1)}$$

Along path I: first $\Delta y \to 0$, then $\Delta x \to 0$.

$$(1) \implies f'(z) = \lim_{\Delta x \to 0} \frac{\Delta x}{\Delta x} \implies \boxed{f'(z) = 1} \qquad \text{— (2)}$$

Along path II: first $\Delta x \to 0$, then $\Delta y \to 0$.

$$(1) \implies f'(z) = \lim_{\Delta y \to 0} \frac{-i\Delta y}{i\Delta y} \implies \boxed{f'(z) = -1} \qquad \text{— (3)}$$

We see that the limits along the two paths are not the same.

$\therefore$ By (2) and (3), $f(z) = \bar{z}$ is **not** a differentiable function.

1.2 Analytic Functions and the Cauchy–Riemann Equations

Analytic function

A complex function $f(z)$ is said to be **analytic** at $z = z_0$ if $f(z)$ is differentiable at $z = z_0$ and also differentiable at every point of some neighbourhood of z_0.

If $f(z)$ is differentiable at each and every point of a region R, then we say that $f(z)$ is **analytic in the region** R.

Cauchy–Riemann (C–R) equations

The necessary condition for a function $f(z)$ to be analytic is given by: if $f(z) = u(x,y) + iv(x,y)$ is analytic, then it satisfies the equations

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y} \quad \text{and} \quad \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}$$

i.e. $\boxed{u_x = v_y}$ and $\boxed{u_y = -v_x}$

Example: $f(z) = z^2$

$f(z) = z^2$ can be written as $f(z) = (x+iy)^2 = (x^2 - y^2) + i\,2xy = u + iv$.

$$u = x^2 - y^2 \qquad v = 2xy$$
$$\frac{\partial u}{\partial x} = 2x \qquad \frac{\partial v}{\partial x} = 2y$$
$$\frac{\partial u}{\partial y} = -2y \qquad \frac{\partial v}{\partial y} = 2x$$

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y} \quad \text{and} \quad \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}$$

The C–R equations are satisfied; $f(z) = z^2$ is an analytic function.

Proof of the C–R equations

Class of 31/08/26

Let $f(z) = u + iv = u(x,y) + i\,v(x,y)$ be an analytic function. Then

$$f'(z) = \lim_{\Delta z \to 0} \frac{f(z+\Delta z) - f(z)}{\Delta z} \qquad \text{— (1)}$$

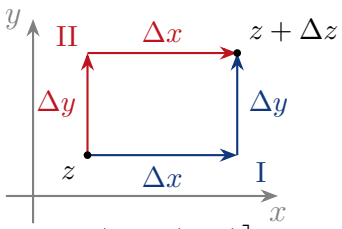
exists and is independent of the path along which $\Delta z \to 0$.

Now $f(z) = u(x,y) + iv(x,y)$ and $f(z+\Delta z) = u(x+\Delta x,\, y+\Delta y) + i\,v(x+\Delta x,\, y+\Delta y)$.

$$\begin{aligned} f(z+\Delta z) - f(z) &= \big[u(x+\Delta x, y+\Delta y) + i\,v(x+\Delta x, y+\Delta y)\big] - \big[u(x,y) + i\,v(x,y)\big] \\ &= \big[u(x+\Delta x, y+\Delta y) - u(x,y)\big] + i\big[v(x+\Delta x, y+\Delta y) - v(x,y)\big] \end{aligned}$$

$$\therefore\ f'(z) = \lim_{(\Delta x + i\Delta y)\to 0} \frac{\big[u(x+\Delta x, y+\Delta y) - u(x,y)\big] + i\big[v(x+\Delta x, y+\Delta y) - v(x,y)\big]}{\Delta x + i\Delta y} \qquad \text{— (2)}$$

Case (i): Along path I — first $\Delta y \to 0$ and then $\Delta x \to 0$.



$$(2) \implies f'(z) = \lim_{\Delta x \to 0} \frac{\big[u(x+\Delta x, y) - u(x,y)\big] + i\big[v(x+\Delta x, y) - v(x,y)\big]}{\Delta x}$$

$$= \lim_{\Delta x \to 0} \frac{u(x+\Delta x, y) - u(x,y)}{\Delta x} + i \lim_{\Delta x \to 0} \frac{v(x+\Delta x, y) - v(x,y)}{\Delta x}$$

$$f'(z) = \frac{\partial u}{\partial x} + i\,\frac{\partial v}{\partial x} \qquad \text{— (3)}$$

Case (ii): Along path II — first $\Delta x \to 0$ and then $\Delta y \to 0$.

$$(2) \implies f'(z) = \lim_{\Delta y \to 0} \frac{\big[u(x, y+\Delta y) - u(x,y)\big] + i\big[v(x, y+\Delta y) - v(x,y)\big]}{i\,\Delta y}$$

$$= \lim_{\Delta y \to 0} \frac{u(x, y+\Delta y) - u(x,y)}{i\,\Delta y} + \lim_{\Delta y \to 0} \frac{v(x, y+\Delta y) - v(x,y)}{\Delta y}$$

$$f'(z) = -i\,\frac{\partial u}{\partial y} + \frac{\partial v}{\partial y} \qquad \text{— (4)} \qquad \left(\because \tfrac{1}{i} = -i\right)$$

Since $f(z)$ is analytic, $f'(z)$ along any path must be the same. $\therefore$ (3) = (4):

$$\frac{\partial u}{\partial x} + i\,\frac{\partial v}{\partial x} = \frac{\partial v}{\partial y} - i\,\frac{\partial u}{\partial y}$$

Equating real and imaginary parts,

C–R equations (Cartesian form)

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y} \quad \text{and} \quad \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x} \quad \text{i.e.} \quad u_x = v_y, \quad u_y = -v_x$$

Hence the C–R equations.

C–R equations in polar form

If we put $z = re^{i\theta}$, we get the complex function $f(z)$ in polar form, $f(z) = u(r,\theta) + i\,v(r,\theta)$. In this case the C–R equations in polar form are:

C–R equations (polar form)

$$\frac{\partial u}{\partial r} = \frac{1}{r}\,\frac{\partial v}{\partial \theta} \quad \text{and} \quad \frac{\partial v}{\partial r} = -\frac{1}{r}\,\frac{\partial u}{\partial \theta}$$

$$\text{i.e.} \quad u_r = \frac{1}{r}\,v_\theta \quad \text{and} \quad v_r = -\frac{1}{r}\,u_\theta$$

1.3 Problems on Analyticity

Problem 1

Check the analyticity of the following functions:

(i) $f(z) = \sin z$ (iii) $f(z) = \cosh z$ (v) $f(z) = z^3$

(ii) $f(z) = xy + iy$ (iv) $f(z) = e^z$ (vi) $f(z) = z^n$

Useful results

$\sin(ix) = i\sinh x$, $\cos(ix) = \cosh x$, $e^{iy} = \cos y + i\sin y$

Solution

(i) Given $f(z) = \sin z$.

$$\begin{aligned} f(z) &= \sin(x+iy) = \sin x \cos(iy) + \cos x \sin(iy) \\ &= \sin x \cosh y + i \cos x \sinh y \\ &= u(x,y) + i\, v(x,y) \end{aligned}$$

$u = \sin x \cosh y$	$v = \cos x \sinh y$
$\dfrac{\partial u}{\partial x} = \cos x \cosh y$	$\dfrac{\partial v}{\partial x} = -\sin x \sinh y$
$\dfrac{\partial u}{\partial y} = \sin x \sinh y$	$\dfrac{\partial v}{\partial y} = \cos x \cosh y$

$$\implies \frac{\partial u}{\partial x} = \frac{\partial v}{\partial y} \quad \text{and} \quad \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}, \qquad \text{i.e. } u_x = v_y \text{ and } u_y = -v_x.$$

C–R equations are satisfied. $\therefore$ $f(z) = \sin z$ is an analytic function.

(ii) Given $f(z) = xy + iy = u + iv$.

$u = xy$	$v = y$	$u_x \neq v_y$ and $u_y \neq -v_x$
$u_x = y$	$v_x = 0$	C–R equations are not satisfied.
$u_y = x$	$v_y = 1$	

$\therefore$ $f(z) = xy + iy$ is **not** an analytic function.

(iii) Given $f(z) = \cosh z$. Let

$$\begin{aligned} \cosh z &= \cos(iz) = \cos\big(i(x+iy)\big) = \cos(ix - y) \\ &= \cos(ix)\cos y + \sin(ix)\sin y \\ &= \cosh x \cos y + i \sinh x \sin y \\ &= u(x,y) + i\, v(x,y) \end{aligned}$$

$u = \cosh x \cos y$	$v = \sinh x \sin y$
$\dfrac{\partial u}{\partial x} = \sinh x \cos y$	$\dfrac{\partial v}{\partial x} = \cosh x \sin y$
$\dfrac{\partial u}{\partial y} = -\cosh x \sin y$	$\dfrac{\partial v}{\partial y} = \sinh x \cos y$

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y} \quad \text{and} \quad \frac{\partial v}{\partial x} = -\frac{\partial u}{\partial y}, \qquad \text{i.e. } u_x = v_y \text{ and } v_x = -u_y.$$

C–R equations are satisfied. $\therefore$ $f(z) = \cosh z$ is an analytic function.

(iv) Given $f(z) = e^z$.

$$f(z) = e^{x+iy} \implies f(z) = e^x e^{iy} = e^x(\cos y + i \sin y) \qquad (\because e^{iy} = \cos y + i \sin y)$$

$$\begin{aligned} u &= e^x \cos y \\ \frac{\partial u}{\partial x} &= e^x \cos y \\ \frac{\partial u}{\partial y} &= -e^x \sin y \end{aligned} \qquad \begin{aligned} v &= e^x \sin y \\ \frac{\partial v}{\partial x} &= e^x \sin y \\ \frac{\partial v}{\partial y} &= e^x \cos y \end{aligned}$$

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y} \quad \text{and} \quad \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}$$

C–R equations are satisfied. $\therefore$ $f(z) = e^z$ is an analytic function.

(v) Given $f(z) = z^3$.

$$\begin{aligned} f(z) &= (x+iy)^3 = x^3 + 3x^2(iy) + 3x(iy)^2 + (iy)^3 \\ &= x^3 + i\,3x^2y - 3xy^2 - i\,y^3 \\ &= (x^3 - 3xy^2) + i\,(3x^2y - y^3) \end{aligned}$$

$$\begin{aligned} u &= x^3 - 3xy^2 \\ \frac{\partial u}{\partial x} &= 3x^2 - 3y^2 \\ \frac{\partial u}{\partial y} &= -6xy \end{aligned} \qquad \begin{aligned} v &= 3x^2y - y^3 \\ \frac{\partial v}{\partial x} &= 6xy \\ \frac{\partial v}{\partial y} &= 3x^2 - 3y^2 \end{aligned}$$

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y} = 3x^2 - 3y^2, \qquad \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x} = -6xy$$

C–R equations are satisfied. $\therefore$ $f(z) = z^3$ is an analytic function.

(vi) Given $f(z) = z^n$. Using the polar form $z = re^{i\theta}$,

$$\begin{aligned} f(z) &= (re^{i\theta})^n = r^n e^{in\theta} = r^n(\cos n\theta + i \sin n\theta) \\ &= r^n \cos n\theta + i\, r^n \sin n\theta = u(r,\theta) + i\, v(r,\theta) \end{aligned}$$

$$\begin{aligned} u &= r^n \cos n\theta \\ \frac{\partial u}{\partial r} &= n\, r^{n-1} \cos n\theta \\ \frac{\partial u}{\partial \theta} &= -n\, r^n \sin n\theta \\ -\frac{1}{r}\frac{\partial u}{\partial \theta} &= n\, r^{n-1} \sin n\theta = \frac{\partial v}{\partial r} \end{aligned} \qquad \begin{aligned} v &= r^n \sin n\theta \\ \frac{\partial v}{\partial r} &= n\, r^{n-1} \sin n\theta \\ \frac{\partial v}{\partial \theta} &= n\, r^n \cos n\theta \\ \frac{1}{r}\frac{\partial v}{\partial \theta} &= n\, r^{n-1} \cos n\theta = \frac{\partial u}{\partial r} \end{aligned}$$

$$\therefore \quad \frac{\partial u}{\partial r} = \frac{1}{r}\frac{\partial v}{\partial \theta} \qquad \text{and} \qquad \frac{\partial v}{\partial r} = -\frac{1}{r}\frac{\partial u}{\partial \theta}$$

C–R equations in polar form are satisfied. $\therefore$ $f(z) = z^n$ is an analytic function.

Problem 2

Class of 02/09/26

Prove that $f(z) = |z|^2$ is differentiable only at the origin $(0,0)$ and is not analytic anywhere.

Solution. Given $f(z) = |z|^2 = x^2 + y^2 = u + iv$, so $u = x^2 + y^2$ and $v = 0$.

$$\frac{\partial u}{\partial x} = 2x \qquad \frac{\partial v}{\partial x} = 0$$
$$\frac{\partial u}{\partial y} = 2y \qquad \frac{\partial v}{\partial y} = 0$$

$\Longrightarrow$ C–R equations are satisfied only when $x = 0$ and $y = 0$. But in the neighbourhood of $(0,0)$, the C–R equations are not satisfied.

$\Longrightarrow$ $f(z) = |z|^2$ is differentiable only at the origin and is not analytic anywhere.

1.4 Properties of Analytic Functions

Property (i)

An analytic function with constant real part is constant.

Proof

Let $f(z) = u + iv$ be an analytic function with $u = c$, a constant. — (1)
Since $f(z) = u + iv$ is analytic, it satisfies the C–R equations

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y} \quad \text{and} \quad \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}$$

$$\Longrightarrow 0 = \frac{\partial v}{\partial y} \quad \text{and} \quad 0 = \frac{\partial v}{\partial x} \qquad (\because u = c) \qquad \text{i.e. } v_x = 0 \text{ and } v_y = 0,$$

which is possible only when $v = k$, a constant.

$$\Longrightarrow f(z) = u + iv = c + ik$$

$f(z)$ is a complex constant.

Note

Similarly we can prove that an analytic function with constant imaginary part is also constant.

Property (ii)

An analytic function with constant modulus is also a constant function.

Proof

Given $f(z) = u + iv$ is an analytic function with constant modulus.

$$\begin{aligned} &\implies |f(z)| = \sqrt{u^2 + v^2} = \text{a constant} \\ &\implies u^2 + v^2 = \text{a constant} = c \text{ (say)} \\ &\implies u^2 + v^2 = c \end{aligned} \qquad \text{— (1)}$$

Differentiating (1) partially w.r.t. x:

$$2uu_x + 2vv_x = 0 \implies uu_x + vv_x = 0 \qquad \text{— (2)}$$

Differentiating (1) partially w.r.t. y:

$$2uu_y + 2vv_y = 0 \implies uu_y + vv_y = 0$$

By the C–R equations ($u_y = -v_x$, $v_y = u_x$):

$$u(-v_x) + v(u_x) = 0 \implies vu_x - uv_x = 0 \qquad \text{— (3)}$$

By (2) and (3),

$$\begin{cases} uu_x + vv_x = 0 \\ vu_x - uv_x = 0 \end{cases} \qquad \begin{vmatrix} u & v \\ v & -u \end{vmatrix} = -u^2 - v^2 = -(u^2 + v^2) = -c \neq 0$$

This is a system of homogeneous equations in u_x and v_x whose determinant value is non-zero.

$$\begin{aligned} &\implies \text{The system possesses only the zero solution} \\ &\implies u_x = 0 \text{ and } v_x = 0 \\ &\implies v_y = 0 \text{ and } u_y = 0 \qquad (\because \text{ by C–R equations}) \end{aligned}$$

$$u_x = u_y = 0 \implies u = c_1, \text{ a constant}; \qquad v_x = v_y = 0 \implies v = c_2, \text{ a constant}$$

$$f(z) = u + iv = c_1 + ic_2$$

$f(z)$ is a complex constant.

Property (iii)

In an analytic function $f(z) = u + iv$, the family of curves $u(x, y) = c_1$ and $v(x, y) = c_2$ intersect orthogonally.

Proof

Given $f(z) = u + iv$ is an analytic function $\implies$ the C–R equations are satisfied:

$$u_x = v_y \qquad \text{and} \qquad u_y = -v_x \qquad \text{— (1)}$$

Consider the family of curves $u(x, y) = c_1$, a constant. Then by the total derivative

$$du = \frac{\partial u}{\partial x}\,dx + \frac{\partial u}{\partial y}\,dy$$

$$\implies 0 = \frac{\partial u}{\partial x}\,dx + \frac{\partial u}{\partial y}\,dy \implies \frac{\partial u}{\partial y}\,dy = -\frac{\partial u}{\partial x}\,dx$$

$$\frac{dy}{dx} = \frac{-\,\partial u/\partial x}{\partial u/\partial y} \qquad \implies \qquad m_1 = -\frac{u_x}{u_y} \qquad -- (2)$$

Now consider the family $v(x, y) = c_2$:

$$dv = \frac{\partial v}{\partial x}\,dx + \frac{\partial v}{\partial y}\,dy$$

$$\implies 0 = \frac{\partial v}{\partial x}\,dx + \frac{\partial v}{\partial y}\,dy \implies \frac{\partial v}{\partial y}\,dy = -\frac{\partial v}{\partial x}\,dx$$

$$\frac{dy}{dx} = -\frac{v_x}{v_y} \qquad \implies \qquad m_2 = -\frac{v_x}{v_y} \qquad -- (3)$$

Then

$$m_1 \cdot m_2 = \left(-\frac{u_x}{u_y}\right) \times \left(-\frac{v_x}{v_y}\right) = \frac{u_x}{u_y} \times \left(\frac{-u_y}{u_x}\right) = -1 \qquad (\because \text{ by C–R equations})$$

$\implies$ The families of curves $u = c_1$ and $v = c_2$ cut orthogonally.

Property (iv)

The real and imaginary parts of an analytic function satisfy Laplace's equation, i.e.

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0 \qquad \text{and} \qquad \frac{\partial^2 v}{\partial x^2} + \frac{\partial^2 v}{\partial y^2} = 0.$$

Proof

By the C–R equations

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y} \quad -- (1) \qquad \text{and} \qquad \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x} \quad -- (2)$$

Differentiate (1) w.r.t. x:

$$\frac{\partial}{\partial x}\left(\frac{\partial u}{\partial x}\right) = \frac{\partial}{\partial x}\left(\frac{\partial v}{\partial y}\right)$$

$$\frac{\partial^2 u}{\partial x^2} = \frac{\partial^2 v}{\partial x\,\partial y}$$

Differentiate (2) w.r.t. y:

$$\frac{\partial}{\partial y}\left(\frac{\partial u}{\partial y}\right) = -\frac{\partial}{\partial y}\left(\frac{\partial v}{\partial x}\right)$$

$$\frac{\partial^2 u}{\partial y^2} = -\frac{\partial^2 v}{\partial y\,\partial x}$$

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = \frac{\partial^2 v}{\partial x\,\partial y} - \frac{\partial^2 v}{\partial y\,\partial x} \qquad \therefore \qquad \boxed{\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0}$$

Similarly, differentiate (1) w.r.t. y:

$$\frac{\partial}{\partial y}\left(\frac{\partial u}{\partial x}\right) = \frac{\partial}{\partial y}\left(\frac{\partial v}{\partial y}\right)$$
$$\frac{\partial^2 u}{\partial y\,\partial x} = \frac{\partial^2 v}{\partial y^2}$$

Similarly, differentiate (2) w.r.t. x:

$$-\frac{\partial}{\partial x}\left(\frac{\partial v}{\partial x}\right) = \frac{\partial}{\partial x}\left(\frac{\partial u}{\partial y}\right)$$
$$-\frac{\partial^2 v}{\partial x^2} = \frac{\partial^2 u}{\partial x\,\partial y}$$

$$\frac{\partial^2 v}{\partial y^2} + \frac{\partial^2 v}{\partial x^2} = \frac{\partial^2 u}{\partial y\,\partial x} - \frac{\partial^2 u}{\partial x\,\partial y} \quad \therefore \quad \boxed{\frac{\partial^2 v}{\partial x^2} + \frac{\partial^2 v}{\partial y^2} = 0}$$

Problem 3

Prove that an analytic function with constant argument is constant.

Proof

Let $f(z) = u(x,y) + iv(x,y) = \rho(x,y)\,e^{ic}$, where the argument c is a constant.

$$u = \rho\cos c$$
$$\frac{\partial u}{\partial x} = \frac{\partial \rho}{\partial x}\cos c$$
$$\frac{\partial u}{\partial y} = \frac{\partial \rho}{\partial y}\cos c$$

$$v = \rho\sin c$$
$$\frac{\partial v}{\partial x} = \frac{\partial \rho}{\partial x}\sin c$$
$$\frac{\partial v}{\partial y} = \frac{\partial \rho}{\partial y}\sin c$$

As $f(z)$ is analytic,

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y} \implies \frac{\partial \rho}{\partial x}\cos c = \frac{\partial \rho}{\partial y}\sin c \implies \rho_x \cos c - \rho_y \sin c = 0 \qquad \text{— (1)}$$

$$\frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x} \implies \frac{\partial \rho}{\partial y}\cos c = -\frac{\partial \rho}{\partial x}\sin c \implies \rho_x \sin c + \rho_y \cos c = 0 \qquad \text{— (2)}$$

$$\begin{pmatrix} \cos c & -\sin c \\ \sin c & \cos c \end{pmatrix}\begin{pmatrix} \rho_x \\ \rho_y \end{pmatrix} = 0, \qquad \begin{vmatrix} \cos c & -\sin c \\ \sin c & \cos c \end{vmatrix} = \cos^2 c + \sin^2 c = 1 \neq 0$$

Since the determinant is non-zero, $\rho_x = \rho_y = 0$. Hence ρ is a constant, and so $f(z) = \rho\,e^{ic}$ is a constant.

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Notes

(i) By the above result (Property (iv)) we can say that u and v are **harmonic functions**, and they are also called **harmonic conjugates** of each other for an analytic function $f(z) = u + iv$.

(ii) For the analytic function $f(z) = u+iv$, from the derivation of the C–R equations, we can write

$$f'(z) = \frac{\partial u}{\partial x} + i\,\frac{\partial v}{\partial x} \quad \text{(or)} \quad f'(z) = -i\,\frac{\partial u}{\partial y} + \frac{\partial v}{\partial y}$$

i.e. $f'(z) = u_x + iv_x$ (or) $f'(z) = -iu_y + v_y$

$$\implies |f'(z)|^2 = u_x^2 + v_x^2 \quad \text{(or)} \quad |f'(z)|^2 = u_y^2 + v_y^2.$$

1.5 Identities for Analytic Functions

Identity (i)

For an analytic function $f(z) = u + iv$,

$$\left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2}\right)|f(z)|^2 = 4\,|f'(z)|^2.$$

Proof

We know that $f(z) = u + iv$ is an analytic function $\implies$ it satisfies the C–R equations.

$$\implies |f(z)|^2 = u^2 + v^2$$

$$\frac{\partial}{\partial x}|f(z)|^2 = \frac{\partial}{\partial x}(u^2 + v^2) = 2uu_x + 2vv_x = 2(uu_x + vv_x)$$

$$\frac{\partial^2}{\partial x^2}|f(z)|^2 = 2\,\frac{\partial}{\partial x}(uu_x + vv_x) = 2\left[u_xu_x + uu_{xx} + v_xv_x + vv_{xx}\right]$$

$$\frac{\partial^2}{\partial x^2}|f(z)|^2 = 2\left[u_x^2 + v_x^2 + uu_{xx} + vv_{xx}\right] \qquad \text{— (1)}$$

Similarly,

$$\frac{\partial^2}{\partial y^2}|f(z)|^2 = 2\left[u_y^2 + v_y^2 + uu_{yy} + vv_{yy}\right] \qquad \text{— (2)}$$

(1) + (2) $\implies$

$$\left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2}\right)|f(z)|^2 = 2\left[(u_x^2 + v_x^2) + (u_y^2 + v_y^2) + u(u_{xx} + u_{yy}) + v(v_{xx} + v_{yy})\right]$$

$$= 2\left[|f'(z)|^2 + |f'(z)|^2 + u(0) + v(0)\right]$$

$(\because$ by Laplace's equation)

$$= 2\left[2|f'(z)|^2\right]$$

$$\boxed{\left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2}\right)|f(z)|^2 = 4\,|f'(z)|^2}$$

Problem 1

Prove that $\left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2}\right)|\operatorname{Re} f(z)|^2 = 2\,|f'(z)|^2$ if $f(z)$ is an analytic function.

Proof

We know that $f(z) = u + iv$ is an analytic function $\implies$ it satisfies the C–R equations.

$$
\begin{aligned}
|\operatorname{Re} f(z)|^2 &= u^2 \\
\frac{\partial}{\partial x}|\operatorname{Re} f(z)|^2 &= \frac{\partial}{\partial x}(u^2) = 2uu_x \\
\frac{\partial^2}{\partial x^2}|\operatorname{Re} f(z)|^2 &= 2\frac{\partial}{\partial x}(uu_x) = 2\big[u_xu_x + uu_{xx}\big] \\
\frac{\partial^2}{\partial x^2}|\operatorname{Re} f(z)|^2 &= 2\big[u_x^2 + uu_{xx}\big] \qquad \text{— (1)}
\end{aligned}
$$

Similarly,

$$
\frac{\partial^2}{\partial y^2}|\operatorname{Re} f(z)|^2 = 2\big[u_y^2 + uu_{yy}\big] \qquad \text{— (2)}
$$

(1) + (2) $\implies$

$$
\begin{aligned}
\left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2}\right)|\operatorname{Re} f(z)|^2 &= 2\Big[(u_x^2 + u_y^2) + u(u_{xx} + u_{yy})\Big] \\
&= 2|f'(z)|^2 + u(0) \\
&\quad (\because\ u_y = -v_x \Rightarrow u_x^2 + u_y^2 = u_x^2 + v_x^2 = |f'(z)|^2)
\end{aligned}
$$

$$
\boxed{\left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2}\right)|\operatorname{Re} f(z)|^2 = 2\,|f'(z)|^2}
$$

Problem 2 (⋆)

If $f(z)$ is analytic, prove that $\left(\dfrac{\partial^2}{\partial x^2} + \dfrac{\partial^2}{\partial y^2}\right)\log_e|f(z)| = 0$.

Proof

Given $f(z) = u + iv$ is an analytic function $\implies$ it satisfies the C–R equations.

$$
|f(z)| = \sqrt{u^2 + v^2} \qquad \implies \qquad \log_e|f(z)| = \tfrac{1}{2}\log_e(u^2 + v^2)
$$

$$
\begin{aligned}
\frac{\partial}{\partial x}\big(\log_e|f(z)|\big) &= \frac{1}{2}\frac{\partial}{\partial x}\big(\log_e(u^2 + v^2)\big) = \frac{1}{2}\left(\frac{1}{u^2 + v^2}\,(2uu_x + 2vv_x)\right) \\
\frac{\partial}{\partial x}\big(\log_e|f(z)|\big) &= \frac{uu_x + vv_x}{u^2 + v^2} \\
\frac{\partial^2}{\partial x^2}\big(\log_e|f(z)|\big) &= \frac{\partial}{\partial x}\left[\frac{uu_x + vv_x}{u^2 + v^2}\right] \\
&= \frac{(u^2 + v^2)(u_xu_x + uu_{xx} + v_xv_x + vv_{xx}) - (uu_x + vv_x)(2uu_x + 2vv_x)}{(u^2 + v^2)^2} \\
&= \frac{(u^2 + v^2)(u_x^2 + uu_{xx} + v_x^2 + vv_{xx}) - 2(u^2u_x^2 + 2uv\,u_xv_x + v^2v_x^2)}{(u^2 + v^2)^2}
\end{aligned}
$$

$$\frac{\partial^2}{\partial x^2}\left(\log_e |f(z)|\right) = \frac{1}{(u^2+v^2)^2}\Big[(u^2+v^2)(u_x^2+v_x^2+uu_{xx}+vv_{xx}) - 2(u^2u_x^2+2uv\,u_xv_x+v^2v_x^2)\Big] \quad \text{— (1)}$$

Similarly,

$$\frac{\partial^2}{\partial y^2}\left(\log_e |f(z)|\right) = \frac{1}{(u^2+v^2)^2}\Big[(u^2+v^2)(u_y^2+v_y^2+uu_{yy}+vv_{yy}) - 2(u^2u_y^2+2uv\,u_yv_y+v^2v_y^2)\Big] \quad \text{— (2)}$$

(1) + (2) $\Longrightarrow$

$$\begin{aligned}
&\left(\frac{\partial^2}{\partial x^2}+\frac{\partial^2}{\partial y^2}\right)\log_e |f(z)| \\
&= \frac{1}{(u^2+v^2)^2}\Big[(u^2+v^2)\big\{(u_x^2+v_x^2)+(u_y^2+v_y^2)+u(u_{xx}+u_{yy})+v(v_{xx}+v_{yy})\big\} \\
&\qquad - 2\big\{u^2(u_x^2+u_y^2)+2uv(u_xv_x+u_yv_y)+v^2(v_x^2+v_y^2)\big\}\Big] \\
&= \frac{1}{(u^2+v^2)^2}\Big[(u^2+v^2)\big\{|f'(z)|^2+|f'(z)|^2+0+0\big\} \\
&\qquad - 2\big\{u^2|f'(z)|^2+0+v^2|f'(z)|^2\big\}\Big] \\
&= \frac{1}{(u^2+v^2)^2}\Big[2(u^2+v^2)|f'(z)|^2-2(u^2+v^2)|f'(z)|^2\Big] = 0
\end{aligned}$$

(Here $u_xv_x+u_yv_y = u_xv_x+(-v_x)(u_x) = 0$ by the C–R equations.)

$$\therefore \quad \boxed{\left(\frac{\partial^2}{\partial x^2}+\frac{\partial^2}{\partial y^2}\right)\log_e |f(z)| = 0}$$

1.6 Construction of Analytic Functions: Milne–Thomson Method

Class of 07/09/26

We can construct the analytic function $f(z) = u + iv$ when either its real part u or its imaginary part v is given.

Milne–Thomson method. Here we follow the procedure of the Milne–Thomson method together with the $f'(z)$ obtained from the derivation of the C–R equations.

Case 1: real part u is given

$$f(z) = \int (\phi_1 - i\,\phi_2)\,dz + c, \qquad \text{where} \quad \phi_1 = \frac{\partial u}{\partial x}(z,0), \quad \phi_2 = \frac{\partial u}{\partial y}(z,0)$$

Case 2: imaginary part v is given

$$f(z) = \int (\psi_1 + i\,\psi_2)\,dz + c, \qquad \text{where} \quad \psi_1 = \frac{\partial v}{\partial y}(z,0), \quad \psi_2 = \frac{\partial v}{\partial x}(z,0)$$

Bernoulli's formula (integration by parts)

$$\int u\,v\,dz = u\,v_1 - u'\,v_2 + u''\,v_3 - \cdots$$

where $v_1, v_2, \ldots$ are the successive integrals of v and $u', u'', \ldots$ the successive derivatives of u.

Problem 1 $(\star)$

Construct the analytic function $f(z) = u + iv$ by the Milne–Thomson method, given $v = e^{-x}(x\cos y + y\sin y)$.

Solution. Given $v = e^{-x}(x\cos y + y\sin y)$.

$$\frac{\partial v}{\partial x} = e^{-x}(\cos y + 0) + (-e^{-x})(x\cos y + y\sin y)$$
$$\frac{\partial v}{\partial x} = e^{-x}\cos y - e^{-x}(x\cos y + y\sin y)$$
$$\psi_2 = \frac{\partial v}{\partial x}(z,0) = e^{-z}\cos 0 - e^{-z}\big(z(1) + 0\big) = e^{-z} - z\,e^{-z}$$

$$\boxed{\psi_2 = (1-z)\,e^{-z}}$$

Now

$$\frac{\partial v}{\partial y} = e^{-x}\big(x(-\sin y) + y\cos y + 1\cdot\sin y\big) = e^{-x}\big((1-x)\sin y + y\cos y\big)$$
$$\psi_1 = \frac{\partial v}{\partial y}(z,0) = e^{-z}\big((1-z)(0) + 0\big)$$

$$\boxed{\psi_1 = 0}$$

$$\therefore\ f(z) = \int (\psi_1 + i\,\psi_2)\,dz = \int \big[0 + i(1-z)e^{-z}\big]dz = i\int (1-z)\,e^{-z}\,dz$$
$$= i\left[(1-z)\frac{e^{-z}}{-1} - (-1)\frac{e^{-z}}{(-1)^2} + 0\right] + c_1$$
$$= i\Big[-(1-z)e^{-z} + e^{-z}\Big] + c_1 = i\Big[-\cancel{e^{-z}} + z\,e^{-z} + \cancel{e^{-z}}\Big] + c_1 = i\,z\,e^{-z} + c_1$$

$f(z) = i\,z\,e^{-z} + c_1$ is the required analytic function.

Problem 2

Construct the analytic function by the Milne–Thomson method where its real part is $u = \dfrac{2\sin 2x}{e^{2y} + e^{-2y} - 2\cos 2x}$.

Solution. Given

$$u = \frac{2\sin 2x}{e^{2y} + e^{-2y} - 2\cos 2x} = \frac{2\sin 2x}{2\cosh 2y - 2\cos 2x} \implies u = \frac{\sin 2x}{\cosh 2y - \cos 2x}$$

Now

$$\begin{aligned}
\frac{\partial u}{\partial x} &= \frac{(\cosh 2y - \cos 2x)(2\cos 2x) - \sin 2x\,(2\sin 2x)}{(\cosh 2y - \cos 2x)^2} \\
&= \frac{2\cosh 2y\cos 2x - 2\cos^2 2x - 2\sin^2 2x}{(\cosh 2y - \cos 2x)^2} \\
\frac{\partial u}{\partial x} &= \frac{2\cosh 2y\cos 2x - 2}{(\cosh 2y - \cos 2x)^2} \\
\phi_1 = \frac{\partial u}{\partial x}(z,0) &= \frac{2\cos 2z - 2}{(1 - \cos 2z)^2} \qquad (\because \cosh 0 = 1) \\
&= \frac{-2\,(1 - \cos 2z)}{(1 - \cos 2z)^2} = \frac{-2}{1 - \cos 2z} \\
&= \frac{-2}{2\sin^2 z} = -\frac{1}{\sin^2 z}
\end{aligned}$$

$$\boxed{\phi_1 = -\operatorname{cosec}^2 z}$$

$$\frac{\partial u}{\partial y} = \frac{(\cosh 2y - \cos 2x)(0) - \sin 2x\,(2\sinh 2y - 0)}{(\cosh 2y - \cos 2x)^2} = \frac{-2\sinh 2y\,\sin 2x}{(\cosh 2y - \cos 2x)^2}$$

$$\phi_2 = \frac{\partial u}{\partial y}(z,0) = 0 \qquad (\because \sinh 0 = 0)$$

$$f(z) = \int (\phi_1 - i\,\phi_2)\,dz = \int \big(-\operatorname{cosec}^2 z - i(0)\big)dz = -\int \operatorname{cosec}^2 z\,dz$$

$$\boxed{f(z) = \cot z + c}$$

Problem 3

Construct the analytic function by the Milne–Thomson method where $u = \dfrac{\sin 2x}{\cos 2x + \cosh 2y}$.

Solution. Given $u = \dfrac{\sin 2x}{\cos 2x + \cosh 2y}$.

$$\begin{aligned}
\frac{\partial u}{\partial x} &= \frac{2\cos 2x\,[\cos 2x + \cosh 2y] - \sin 2x\,(-2\sin 2x)}{(\cos 2x + \cosh 2y)^2} \\
&= \frac{2\cos 2x\,[\cos 2x + \cosh 2y] + 2\sin^2 2x}{(\cos 2x + \cosh 2y)^2} \\
&= \frac{2\cos^2 2x + 2\cos 2x\cosh 2y + 2\sin^2 2x}{(\cos 2x + \cosh 2y)^2} \\
&= \frac{2\big[\cos^2 2x + \sin^2 2x\big] + 2\cos 2x\cosh 2y}{(\cos 2x + \cosh 2y)^2}
\end{aligned}$$

$$\frac{\partial u}{\partial x} = \frac{2 + 2\cos 2x \cosh 2y}{(\cos 2x + \cosh 2y)^2}$$

$$\frac{\partial u}{\partial x}(z,0) = \frac{2 + 2\cos 2z \cosh 0}{[\cos 2z + \cosh 0]^2} = \frac{2 + 2\cos 2z}{[\cos 2z + 1]^2}$$

$$= \frac{2\,[1 + \cancel{\cos 2z}]}{[1 + \cos 2z]^{\cancel{2}}} = \frac{2}{1 + \cos 2z}$$

$$= \frac{\cancel{2}}{\cancel{2}\cos^2 z} = \frac{1}{\cos^2 z} = \sec^2 z$$

$$\boxed{\phi_1 = \tfrac{\partial u}{\partial x}(z,0) = \sec^2 z}$$

$$\frac{\partial u}{\partial y} = \frac{(0)(\cos 2x + \cosh 2y) - (0 + 2\sinh 2y)(\sin 2x)}{(\cos 2x + \cosh 2y)^2} = \frac{-2\sin 2x \sinh 2y}{(\cos 2x + \cosh 2y)^2}$$

$$\frac{\partial u}{\partial y}(z,0) = 0 \qquad (\because \sinh 0 = 0)$$

$$\boxed{\phi_2 = \tfrac{\partial u}{\partial y}(z,0) = 0}$$

$$f(z) = \int (\phi_1 - i\,\phi_2)\,dz = \int \big(\sec^2 z - i(0)\big)dz$$

$$\boxed{f(z) = \tan z + c}$$

Note on the question

The handwritten notes write the denominator as $\cos 2x + \cos 2y$. With $\cos 2y$ the given u is not harmonic, so it cannot be the real part of any analytic function. The standard form of this question has $\cosh 2y$, and then $\mathrm{Re}(\tan z) = \dfrac{\sin 2x}{\cos 2x + \cosh 2y}$ exactly. (Both versions give the same ϕ_1, ϕ_2 at $(z,0)$, which is why the method still produced $\tan z$.)

Problem 4

Construct the analytic function $f(z) = u + iv$ by the Milne–Thomson method, given $v = \dfrac{x-y}{x^2+y^2}$.

Solution. Given $v = \dfrac{x-y}{x^2+y^2}$.

$$\frac{\partial v}{\partial x} = \frac{(1)(x^2+y^2) - (x-y)(2x)}{(x^2+y^2)^2} = \frac{x^2+y^2-2x^2+2xy}{(x^2+y^2)^2} = \frac{-x^2+y^2+2xy}{(x^2+y^2)^2}$$

$$\frac{\partial v}{\partial x}(z,0) = \frac{-z^2}{(z^2)^2}$$

$$\boxed{\psi_2 = -\frac{1}{z^2}}$$

$$\frac{\partial v}{\partial y} = \frac{(-1)(x^2+y^2) - (x-y)(2y)}{(x^2+y^2)^2} = \frac{-x^2-y^2-2xy+2y^2}{(x^2+y^2)^2} = \frac{-x^2+y^2-2xy}{(x^2+y^2)^2}$$

$$\frac{\partial v}{\partial y}(z,0) = \frac{-z^2}{(z^2)^2}$$

$$\boxed{\psi_1 = -\frac{1}{z^2}}$$

$$f(z) = \int (\psi_1 + i\,\psi_2)\,dz = \int \left[\left(-\frac{1}{z^2}\right) + i\left(-\frac{1}{z^2}\right)\right] dz = -(1+i)\int z^{-2}\,dz$$
$$= -(1+i)\left(-\frac{1}{z}\right) + c$$

$$\boxed{f(z) = \frac{1+i}{z} + c}$$

Problems 5 & 6

Construct the analytic function $f(z) = u + iv$ by the Milne–Thomson method, given

(5) $v = e^x(x\sin y + y\cos y)$

(6) $u = \frac{1}{2}\log_e(x^2+y^2)$

5) Solution. Given $v = e^x(x\sin y + y\cos y)$.

$$\frac{\partial v}{\partial y} = e^x\left[x\cos y + (1)\cos y - y\sin y\right] = e^x\left[x\cos y + \cos y - y\sin y\right]$$
$$\psi_1 = \frac{\partial v}{\partial y}(z,0) = e^z\big(z(1) + (1) - 0\big) = e^z(1+z)$$

$$\frac{\partial v}{\partial x} = e^x(\sin y) + e^x(x\sin y + y\cos y)$$
$$\psi_2 = \frac{\partial v}{\partial x}(z,0) = e^z(0) + e^z\big(z(0)+0\big) = 0$$

$$f(z) = \int (\psi_1 + i\,\psi_2)\,dz = \int \left[e^z + z\,e^z\right]dz = e^z + \left[z\,e^z - e^z\right] + c$$

$$\boxed{f(z) = z\,e^z + c}$$

6) Solution. Given $u = \frac{1}{2}\log_e(x^2+y^2)$.

$$\frac{\partial u}{\partial x} = \frac{1}{2}\left(\frac{1}{x^2+y^2}\right)(2x), \qquad \phi_1 = \frac{\partial u}{\partial x}(z,0) = \frac{1}{2}\left[\frac{1}{z^2+0}\times 2z\right] = \frac{1}{z}$$
$$\frac{\partial u}{\partial y} = \frac{1}{2}\left(\frac{1}{x^2+y^2}\right)(2y), \qquad \phi_2 = \frac{\partial u}{\partial y}(z,0) = \frac{1}{2z^2}\big(2(0)\big) = 0$$

$$f(z) = \int (\phi_1 - i\,\phi_2)\,dz = \int \frac{1}{z}\,dz$$

$$\boxed{f(z) = \log z + c}$$

When a linear combination of u and v is given

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Problem 1

Construct the analytic function $f(z)$ by the Milne–Thomson method where

$$2u + v = e^{2x}\big[(2x+y)\cos 2y + (x-2y)\sin 2y\big].$$

Solution. Given $2u + v = e^{2x}\big[(2x+y)\cos 2y + (x-2y)\sin 2y\big]$.

Let $f(z) = u + iv$ be the required analytic function. — (1)

$$(1)\times 2 \implies 2f(z) = 2u + i\,2v$$
$$(1)\times(-i) \implies -i\,f(z) = -iu + v$$

Adding the above two equations,

$$(2-i)f(z) = (2u+v) + i(2v-u) \qquad \Longrightarrow \qquad F(z) = U + iV \quad — (2)$$

where $F(z) = (2-i)f(z)$, $U = 2u + v$ and $V = 2v - u$.

So in the above analytic function $F(z) = U + iV$, we have its real part

$$U = 2u + v = e^{2x}\big[(2x+y)\cos 2y + (x-2y)\sin 2y\big]$$

$$\frac{\partial U}{\partial x} = 2e^{2x}\big[(2x+y)\cos 2y + (x-2y)\sin 2y\big] + e^{2x}\big[2\cos 2y + \sin 2y\big]$$

$$\phi_1 = \frac{\partial U}{\partial x}(z,0) = 2e^{2z}[2z] + e^{2z}[2]$$

$$\boxed{\phi_1 = 2e^{2z}(2z+1)}$$

$$\frac{\partial U}{\partial y} = e^{2x}\big[(2x+y)(-\sin 2y)(2) + \cos 2y + (x-2y)(2\cos 2y) + (-2)\sin 2y\big]$$

$$\phi_2 = \frac{\partial U}{\partial y}(z,0) = e^{2z}\big[1 + z(2)\big]$$

$$\boxed{\phi_2 = (2z+1)\,e^{2z}}$$

$$\therefore\ F(z) = \int (\phi_1 - i\,\phi_2)\,dz = \int \Big[2e^{2z}(2z+1) - i(2z+1)e^{2z}\Big]dz + c$$

$$= \int (2-i)\,(2z+1)\,\underset{u}{e^{2z}}\,\underset{v}{dz} + c = (2-i)\int (2z+1)\,e^{2z}\,dz + c$$

$$= (2-i)\left[(2z+1)\left(\frac{e^{2z}}{2}\right) - (2)\left(\frac{e^{2z}}{4}\right) + 0\right] + c$$

$$= \frac{2-i}{2}\Big[e^{2z}\big(2z+\cancel{1}-\cancel{1}\big)\Big] + c = \frac{2-i}{2}\big[2z\,e^{2z}\big] + c$$

$$F(z) = (2-i)\,z\,e^{2z} + c$$

But $F(z) = (2-i)f(z)$:

$$(2-i)f(z) = (2-i)\,z\,e^{2z} + c \implies f(z) = z\,e^{2z} + \frac{c}{2-i}$$

$$\boxed{f(z) = z\,e^{2z} + k, \quad \text{where } k \text{ is a complex constant.}}$$

Problem 2

Construct the analytic function $f(z) = u + iv$, given $3u + 2v = y^2 - x^2 + 16xy$.

Solution. Given $3u + 2v = y^2 - x^2 + 16xy$.

Let $f(z) = u + iv$ be an analytic function. — (1)

$$(1) \times 3 \implies 3f(z) = 3u + i\,3v$$
$$(1) \times (-2i) \implies -2i\,f(z) = -2iu + 2v$$

Add $\implies (3-2i)f(z) = (3u+2v)+i(3v-2u)$, i.e. $F(z) = U+iV$, where $F(z) = (3-2i)f(z)$ and $U = 3u + 2v$ is given:

$$U = y^2 - x^2 + 16xy$$

$$\frac{\partial U}{\partial x} = -2x + 16y \qquad \frac{\partial U}{\partial y} = 2y + 16x$$
$$\phi_1 = \frac{\partial U}{\partial x}(z, 0) = -2z \qquad \phi_2 = \frac{\partial U}{\partial y}(z, 0) = 16z$$

$$F(z) = \int (\phi_1 - i\,\phi_2)\,dz = \int \phi_1\,dz - i\int \phi_2\,dz + c = \int -2z\,dz - i\int 16z\,dz + c$$
$$= (-2 - 16i)\int z\,dz + c = (-2 - 16i)\left(\frac{z^2}{2}\right) + c$$
$$F(z) = -(1 + 8i)\,z^2 + c$$

$$(3 - 2i)f(z) = -(1 + 8i)\,z^2 + c$$
$$f(z) = -\frac{1 + 8i}{3 - 2i}\,z^2 + \frac{c}{3 - 2i} = -z^2\left[\frac{1 + 8i}{3 - 2i} \times \frac{3 + 2i}{3 + 2i}\right] + k$$
$$= -z^2\left[\frac{3 + 2i + 24i - 16}{9 + 4}\right] + k = -z^2\left[\frac{26i - 13}{13}\right] + k$$

$$f(z) = (1 - 2i)\,z^2 + k$$

Problem 3

Construct the analytic function $f(z) = u + iv$ where

(i) $u + v = \dfrac{x}{x^2 + y^2}$

(ii) $u - v = \dfrac{\cos x + \sin x - e^{-y}}{2\cos x - (e^y + e^{-y})}$, given $f\left(\frac{\pi}{2}\right) = 0$.

(i) Solution. Given $u + v = \dfrac{x}{x^2 + y^2}$.

Let $f(z) = u + iv$ be an analytic function. — (1)

$$(1) \times (-i) \implies -i\,f(z) = -iu + v \qquad — (2)$$

Add (1) and (2):

$$(1-i)f(z) = (u+v) + i(v-u) \qquad \Longrightarrow \qquad F(z) = U + iV$$

where $F(z) = (1-i)f(z)$ and $U = u + v = \dfrac{x}{x^2+y^2}$.

$$\frac{\partial U}{\partial x} = \frac{(x^2+y^2)(1) - x(2x)}{(x^2+y^2)^2} = \frac{(x^2+y^2) - 2x^2}{(x^2+y^2)^2}$$

$$\frac{\partial U}{\partial x}(z,0) = \frac{z^2 - 2z^2}{(z^2)^2} = \frac{-z^2}{z^4}$$

$$\boxed{\phi_1 = \tfrac{\partial U}{\partial x}(z,0) = -\frac{1}{z^2}}$$

$$\frac{\partial U}{\partial y} = \frac{(x^2+y^2)(0) - x(2y)}{(x^2+y^2)^2} = \frac{-2xy}{(x^2+y^2)^2}, \qquad \frac{\partial U}{\partial y}(z,0) = 0$$

$$\boxed{\phi_2 = \tfrac{\partial U}{\partial y}(z,0) = 0}$$

$$F(z) = \int (\phi_1 - i\,\phi_2)\,dz = \int \left(-\frac{1}{z^2}\right) dz + c \quad \Longrightarrow \quad F(z) = \frac{1}{z} + c$$

$$(1-i)f(z) = \frac{1}{z} + c \quad \Longrightarrow \quad f(z) = \frac{1}{z(1-i)} + \frac{c}{1-i} = \frac{1+i}{z(1-i)(1+i)} + k$$

$$\boxed{f(z) = \frac{1+i}{2z} + k}$$

(ii) Solution. Given $u - v = \dfrac{\cos x + \sin x - e^{-y}}{2\cos x - (e^y + e^{-y})}$ and $f(\pi/2) = 0$.

Let $f(z) = u + iv$ be an analytic function. — (1)

$$(1) \times i \implies i\,f(z) = iu - v \qquad \text{— (2)}$$

Add:

$$f(z) + i\,f(z) = (u-v) + i(v+u) \implies (1+i)f(z) = (u-v) + i(u+v) \implies F(z) = U + iV,$$

where $F(z) = (1+i)f(z)$ and

$$U = u - v = \frac{\cos x + \sin x - e^{-y}}{2\cos x - (e^y + e^{-y})} = \frac{\cos x + \sin x - e^{-y}}{2(\cos x - \cosh y)} \qquad (\because\ e^y + e^{-y} = 2\cosh y)$$

$$\frac{\partial U}{\partial x} = \frac{2(\cos x - \cosh y)(-\sin x + \cos x) - (\cos x + \sin x - e^{-y})(-2\sin x)}{\big(2(\cos x - \cosh y)\big)^2}$$

$$\phi_1 = \frac{\partial U}{\partial x}(z,0) = \frac{(2\cos z - 2)(-\sin z + \cos z) - (\cos z + \sin z - 1)(-2\sin z)}{4(\cos z - 1)^2}$$

$$= \frac{2\left[\cos^2 z - \cos z - \sin z\cos z + \sin z + \sin z\cos z + \sin^2 z - \sin z\right]}{4(\cos z - 1)^2}$$

$$= \frac{2(1-\cos z)}{4(1-\cos z)^2} = \frac{1}{2(1-\cos z)} = \frac{1}{4\sin^2(z/2)}$$

$$\boxed{\phi_1 = \tfrac{1}{4}\operatorname{cosec}^2\left(\tfrac{z}{2}\right)}$$

$$\frac{\partial U}{\partial y} = \frac{2(\cos x - \cosh y)(e^{-y}) - (\cos x + \sin x - e^{-y})(-2\sinh y)}{4(\cos x - \cosh y)^2}$$

$$\phi_2 = \frac{\partial U}{\partial y}(z,0) = \frac{2(\cos z - 1)(1) - 0}{4(\cos z - 1)^2} = \frac{1}{2(\cos z - 1)} = -\frac{1}{2(1-\cos z)} = -\frac{1}{4\sin^2(z/2)}$$

$$\boxed{\phi_2 = -\tfrac{1}{4}\operatorname{cosec}^2\left(\tfrac{z}{2}\right)}$$

$$\begin{aligned}
F(z) &= \int (\phi_1 - i\,\phi_2)\,dz = \int \left(\tfrac{1}{4}\operatorname{cosec}^2\left(\tfrac{z}{2}\right) + i\,\tfrac{1}{4}\operatorname{cosec}^2\left(\tfrac{z}{2}\right)\right) dz \\
&= \left(\frac{1+i}{4}\right)\int \operatorname{cosec}^2\left(\tfrac{z}{2}\right) dz \\
&= \left(\frac{1+i}{4}\right)\left[\frac{-\cot(z/2)}{1/2}\right] + c \\
F(z) &= -\left(\frac{1+i}{2}\right)\cot\left(\frac{z}{2}\right) + c \\
(1+i)f(z) &= -\left(\frac{1+i}{2}\right)\cot\left(\frac{z}{2}\right) + c \\
\implies \quad f(z) &= -\frac{1}{2}\cot\left(\frac{z}{2}\right) + k
\end{aligned}$$

Given $f(\pi/2) = 0$:

$$-\tfrac{1}{2}\cot\left(\tfrac{\pi}{4}\right) + k = 0 \implies -\tfrac{1}{2}(1) + k = 0 \implies k = \tfrac{1}{2}$$

$$\boxed{f(z) = \frac{1}{2}\left[1 - \cot\left(\frac{z}{2}\right)\right]}$$

Check

In the handwritten notes the signs of ϕ_1 and ϕ_2 were both interchanged, which led to $f(z) = \frac{1}{2}\big[\cot(z/2) - 1\big]$. With the signs as worked above, $\operatorname{Re} f - \operatorname{Im} f$ reproduces the given $u - v$ exactly, so the answer is $f(z) = \frac{1}{2}\big[1 - \cot(z/2)\big]$.

1.7 Problems on Analytic and Harmonic Functions

Class of 10/09/26

Problem 1

If $f(z) = u + iv$ is an analytic function, prove that (a) $v - iu$ is also an analytic function; (b) $v + iu$ is not an analytic function.

Solution. $f(z) = u + iv$ is an analytic function, so its real part u and imaginary part v must satisfy the Cauchy–Riemann (C–R) equations:

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y} \implies u_x = v_y \quad \text{— (1)} \qquad \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x} \implies u_y = -v_x \quad \text{— (2)}$$

(a) Let $F(z) = v - iu = -i(u + iv)$. Since $f(z) = u + iv$,

$$F(z) = -i\, f(z)$$

which is a constant multiple of an analytic function.

$\therefore$ $v - iu$ is an analytic function.

(b) Let $F(z) = v + iu = U + iV$, with $U = v$ and $V = u$. If F were analytic, the C–R equations $U_x = V_y$, $U_y = -V_x$ would give

$$v_x = u_y \quad \text{— (3)} \qquad \text{and} \qquad v_y = -u_x \quad \text{— (4)}$$

From (2): $u_y = -v_x$, and by (3)

$$v_x = -v_x$$
$$2v_x = 0 \implies v_x = 0$$

From (1): $u_x = v_y$, and by (4)

$$v_y = -v_y$$
$$2v_y = 0 \implies v_y = 0$$

This would force u and v to be constants. But u and v are not constants, so the C–R equations are not satisfied.

$\therefore$ $v + iu$ is not an analytic function.

Problem 2

Find the points at which the C–R equations are satisfied by the function $xy^2 + i\, yx^2$.

Solution. Let $f(z) = u + iv$. Given $f(z) = xy^2 + i\, yx^2$, so $u = xy^2$, $v = x^2y$.

$$\frac{\partial u}{\partial x} = y^2$$
$$\frac{\partial u}{\partial y} = 2xy$$

$$\frac{\partial v}{\partial x} = 2xy$$
$$\frac{\partial v}{\partial y} = x^2$$

By the C–R equations

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y} \implies y^2 = x^2 \quad \text{— (1)}$$
$$\frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x} \implies 2xy = -2xy \implies 4xy = 0 \implies xy = 0 \quad \text{— (2)}$$

From (2), $x = 0$ or $y = 0$. If $x = 0$, (1) gives $y = 0$; if $y = 0$, (1) gives $x = 0$.

The C–R equations are satisfied only at the point $(0, 0)$, i.e. at $z = 0$.
$\therefore$ The function is not analytic anywhere.

Problem 3

Determine the value of p such that $f(z) = \frac{1}{2}\log(x^2+y^2) + i\tan^{-1}\left(\frac{px}{y}\right)$ is an analytic function.

Solution. Given $f(z)$ is analytic, with $u = \frac{1}{2}\log(x^2+y^2)$ and $v = \tan^{-1}\left(\frac{px}{y}\right)$.

$$\frac{\partial u}{\partial x} = \frac{1}{2}\cdot\frac{2x}{x^2+y^2} = \frac{x}{x^2+y^2}$$
$$\frac{\partial u}{\partial y} = \frac{1}{2}\cdot\frac{2y}{x^2+y^2} = \frac{y}{x^2+y^2}$$

$$\frac{\partial v}{\partial x} = \frac{1}{1+\frac{p^2x^2}{y^2}}\cdot\frac{p}{y} = \frac{py}{y^2+p^2x^2}$$
$$\frac{\partial v}{\partial y} = \frac{1}{1+\frac{p^2x^2}{y^2}}\left(\frac{-px}{y^2}\right) = \frac{-px}{y^2+p^2x^2}$$

By the C–R equations,

$$\begin{aligned}\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y} &\implies \frac{\cancel{x}}{x^2+y^2} = \frac{-p\cancel{x}}{y^2+p^2x^2}\\ &\implies y^2+p^2x^2 = -p(x^2+y^2) = -px^2-py^2\\ &\implies p^2x^2+px^2+y^2+py^2 = 0\\ &\implies x^2(p^2+p)+y^2(1+p) = 0\end{aligned}$$

$p^2+p = 0 \Rightarrow p(p+1) = 0 \Rightarrow p = 0$ or $p = -1$

$$1+p = 0 \implies p = -1$$

Check the second equation with $p = -1$:

$$\frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x} \implies \frac{y}{x^2+y^2} = -\frac{(-1)y}{y^2+(-1)^2x^2} = \frac{y}{y^2+x^2} \quad \checkmark$$

C–R equations are satisfied; $p = -1$ makes $f(z)$ an analytic function.

Problem 4

If $f(z) = u + iv = \cosh z$ is an analytic function, show that the curves $u = \text{const}$ and $v = \text{const}$ are orthogonal.

Solution. Given $f(z) = u + iv = \cosh z$.

$$\cosh z = \cosh(x+iy) = \cosh x\cos y + i\sinh x\sin y$$

$$u = \cosh x\cos y$$
$$\frac{\partial u}{\partial x} = \sinh x\cos y$$
$$\frac{\partial u}{\partial y} = -\cosh x\sin y$$

$$v = \sinh x\sin y$$
$$\frac{\partial v}{\partial x} = \cosh x\sin y$$
$$\frac{\partial v}{\partial y} = \sinh x\cos y$$

By the C–R equations:

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y}: \ \sinh x\cos y = \sinh x\cos y; \qquad \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}: \ -\cosh x\sin y = -\cosh x\sin y.$$

Consider the family of curves $u(x,y) = c_1$:

$$0 = du = \frac{\partial u}{\partial x}\,dx + \frac{\partial u}{\partial y}\,dy \implies \frac{\partial u}{\partial y}\,dy = -\frac{\partial u}{\partial x}\,dx \implies \frac{dy}{dx} = \frac{-\,\partial u/\partial x}{\partial u/\partial y} \implies m_1 = -\frac{u_x}{u_y}$$

and the family $v(x,y) = c_2$:

$$0 = dv = \frac{\partial v}{\partial x}\,dx + \frac{\partial v}{\partial y}\,dy \implies \frac{\partial v}{\partial y}\,dy = -\frac{\partial v}{\partial x}\,dx \implies \frac{dy}{dx} = -\frac{v_x}{v_y} \implies m_2 = -\frac{v_x}{v_y}$$

$$m_1 \cdot m_2 = \left(-\frac{u_x}{u_y}\right)\left(-\frac{v_x}{v_y}\right) = \frac{u_x\,v_x}{u_y\,v_y} = \frac{v_y \cdot v_x}{(-v_x)\cdot v_y} = -1 \qquad (\because\ u_x = v_y,\ u_y = -v_x)$$

$m_1 \cdot m_2 = -1$: the two families of curves $u = \text{const}$ and $v = \text{const}$ cut each other at right angles.

Problem 5

Find a, b if $f(z) = (x^2 - 2xy + ay^2) + i(bx^2 - y^2 + 2xy)$ is an analytic function, given a and b are real, and also find $f(z)$ in terms of z.

Solution. Given $f(z) = u + iv = (x^2 - 2xy + ay^2) + i(bx^2 - y^2 + 2xy)$.

$$u = x^2 - 2xy + ay^2 \qquad\qquad v = bx^2 - y^2 + 2xy$$
$$\frac{\partial u}{\partial x} = 2x - 2y \qquad\qquad \frac{\partial v}{\partial x} = 2bx + 2y$$
$$\frac{\partial u}{\partial y} = -2x + 2ay \qquad\qquad \frac{\partial v}{\partial y} = -2y + 2x$$

By the C–R equations,

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y} \implies 2x - 2y = 2x - 2y$$
$$\frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x} \implies -2x + 2ay = -(2bx + 2y) = -2bx - 2y$$

Comparing the x and y terms:

$$-2 = -2b \implies \boxed{b = 1} \qquad\qquad 2a = -2 \implies \boxed{a = -1}$$

Substituting a in u: $u = x^2 - 2xy - y^2$.

$$\frac{\partial u}{\partial x} = 2x - 2y, \qquad \phi_1 = \frac{\partial u}{\partial x}(z,0) = 2(z) - 2(0) = 2z$$
$$\frac{\partial u}{\partial y} = -2x - 2y, \qquad \phi_2 = \frac{\partial u}{\partial y}(z,0) = -2(z) - 2(0) = -2z$$

$$f(z) = \int (\phi_1 - i\,\phi_2)\,dz = \int \big(2z - i(-2z)\big)dz = \int (2z + 2iz)\,dz$$
$$= (2 + 2i)\int z\,dz = (2 + 2i)\left(\frac{z^2}{2}\right) + c$$

$$f(z) = (1 + i)\,z^2 + c$$

Problem 6

If u is a harmonic function, show that u^2 is not a harmonic function unless u is a constant.

Solution. Given u is a harmonic function:

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0 \qquad \text{— (1)}$$

To check whether u^2 is harmonic, compute $\left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2}\right)(u^2)$.

Differentiating w.r.t. x:

$$\frac{\partial}{\partial x}(u^2) = 2u\frac{\partial u}{\partial x}$$
$$\frac{\partial^2}{\partial x^2}(u^2) = 2u\frac{\partial^2 u}{\partial x^2} + 2\left(\frac{\partial u}{\partial x}\right)^2 \qquad \text{— (2)}$$

Differentiating w.r.t. y:

$$\frac{\partial}{\partial y}(u^2) = 2u\frac{\partial u}{\partial y}$$
$$\frac{\partial^2}{\partial y^2}(u^2) = 2u\frac{\partial^2 u}{\partial y^2} + 2\left(\frac{\partial u}{\partial y}\right)^2 \qquad \text{— (3)}$$

(2) + (3) $\Longrightarrow$

$$\frac{\partial^2}{\partial x^2}(u^2) + \frac{\partial^2}{\partial y^2}(u^2) = 2u\left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2}\right) + 2\left[\left(\frac{\partial u}{\partial x}\right)^2 + \left(\frac{\partial u}{\partial y}\right)^2\right]$$

Substituting (1):

$$\left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2}\right)u^2 = 0 + 2\left[\left(\frac{\partial u}{\partial x}\right)^2 + \left(\frac{\partial u}{\partial y}\right)^2\right]$$

For u^2 to be harmonic we need $\left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2}\right)(u^2) = 0$:

$$\Longrightarrow\ 2\left[\left(\frac{\partial u}{\partial x}\right)^2 + \left(\frac{\partial u}{\partial y}\right)^2\right] = 0 \ \Longrightarrow\ \frac{\partial u}{\partial x} = 0 \text{ and } \frac{\partial u}{\partial y} = 0,$$

so u must be a constant function.

u^2 is not harmonic unless u is a constant.

Problem 7

Find whether $u = y + e^x \cos y$ can be the real part of an analytic function.

Solution. Given $u = y + e^x \cos y$.

$$\frac{\partial u}{\partial x} = e^x \cos y$$
$$\frac{\partial^2 u}{\partial x^2} = e^x \cos y$$

$$\frac{\partial u}{\partial y} = 1 + e^x(-\sin y) = 1 - e^x \sin y$$
$$\frac{\partial^2 u}{\partial y^2} = -e^x \cos y$$

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = e^x \cos y + (-e^x \cos y) = 0$$

u satisfies Laplace's equation, so u is harmonic.

$\therefore$ $u = y + e^x \cos y$ can be the real part of an analytic function.

Problem 8

If $u = x^2 - y^2$ and $v = \dfrac{-2y}{x^2+y^2}$, prove that u and v are harmonic functions, but they are not harmonic conjugates of each other.

Solution. Given $u = x^2 - y^2$ and $v = \dfrac{-2y}{x^2+y^2}$.

$$\frac{\partial u}{\partial x} = 2x$$
$$\frac{\partial^2 u}{\partial x^2} = 2$$
$$\frac{\partial u}{\partial y} = -2y$$
$$\frac{\partial^2 u}{\partial y^2} = -2$$

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 2 + (-2) = 0$$

$\therefore$ u is a harmonic function.

$$\frac{\partial v}{\partial x} = -2y \cdot \frac{-1}{(x^2+y^2)^2}(2x) = \frac{4xy}{(x^2+y^2)^2}$$
$$\frac{\partial^2 v}{\partial x^2} = \frac{(x^2+y^2)^2(4y) - 4xy \cdot 2(x^2+y^2)(2x)}{(x^2+y^2)^4}$$
$$= \frac{4y^3 - 12x^2y}{(x^2+y^2)^3}$$
$$\frac{\partial v}{\partial y} = \frac{(x^2+y^2)(-2) - (-2y)(2y)}{(x^2+y^2)^2} = \frac{-2x^2+2y^2}{(x^2+y^2)^2}$$
$$\frac{\partial^2 v}{\partial y^2} = \frac{(x^2+y^2)^2(4y) - (-2x^2+2y^2) \cdot 2(x^2+y^2)(2y)}{(x^2+y^2)^4}$$
$$= \frac{12x^2y - 4y^3}{(x^2+y^2)^3}$$

$$\frac{\partial^2 v}{\partial x^2} + \frac{\partial^2 v}{\partial y^2} = \frac{4y^3 - 12x^2y + 12x^2y - 4y^3}{(x^2+y^2)^3} = 0$$

$\therefore$ v is also a harmonic function.

To check whether u and v are harmonic conjugates, they must satisfy the C–R equations $\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y}$ and $\frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}$. But

$$\frac{\partial u}{\partial x} = 2x \neq \frac{2y^2 - 2x^2}{(x^2+y^2)^2} = \frac{\partial v}{\partial y} \qquad \therefore \frac{\partial u}{\partial x} \neq \frac{\partial v}{\partial y}$$

C–R equations are not satisfied; u and v are not harmonic conjugates.

Problem 9

If $f(z) = u(r,\theta) + i\,v(r,\theta)$ is an analytic function, prove that

$$r^2\frac{\partial^2 u}{\partial r^2} + r\frac{\partial u}{\partial r} + \frac{\partial^2 u}{\partial \theta^2} = 0.$$

(This equation is called **Laplace's equation in polar form**.)

Proof

Given $f(z) = u + iv$ is analytic, so the polar C–R equations hold:

$$\frac{\partial u}{\partial r} = \frac{1}{r}\frac{\partial v}{\partial \theta} \qquad \frac{\partial v}{\partial \theta} = r\frac{\partial u}{\partial r}$$
$$\frac{\partial u}{\partial \theta} = -r\frac{\partial v}{\partial r} \qquad \frac{\partial v}{\partial r} = -\frac{1}{r}\frac{\partial u}{\partial \theta}$$

$$\frac{\partial^2 v}{\partial r\,\partial \theta} = \frac{\partial}{\partial r}\left(r\frac{\partial u}{\partial r}\right) = (1)\frac{\partial u}{\partial r} + r\frac{\partial^2 u}{\partial r^2}$$
$$\frac{\partial^2 v}{\partial \theta\,\partial r} = \frac{\partial}{\partial \theta}\left(-\frac{1}{r}\frac{\partial u}{\partial \theta}\right) = -\frac{1}{r}\frac{\partial^2 u}{\partial \theta^2}$$

Since $\dfrac{\partial^2 v}{\partial r\,\partial \theta} = \dfrac{\partial^2 v}{\partial \theta\,\partial r}$,

$$\frac{\partial u}{\partial r} + r\frac{\partial^2 u}{\partial r^2} = -\frac{1}{r}\frac{\partial^2 u}{\partial \theta^2}$$
$$r\frac{\partial u}{\partial r} + r^2\frac{\partial^2 u}{\partial r^2} = -\frac{\partial^2 u}{\partial \theta^2}$$

$$\boxed{r^2\frac{\partial^2 u}{\partial r^2} + r\frac{\partial u}{\partial r} + \frac{\partial^2 u}{\partial \theta^2} = 0}$$

Hence proved.

2 Problems on Analytic Functions

Practice Questions

2.1 Problems in Analytic Functions

Problems

1. If $f(z) = u + iv$ is an analytic function, prove that
   (a) $v - iu$ is also an analytic function.
   (b) $v + iu$ is not an analytic function.
2. Find the points at which C-R equations are satisfied by the function $xy^2 + iyx^2$.
3. Determine the value of p such that
   $$f(z) = \tfrac{1}{2}\log(x^2 + y^2) + i\tan^{-1}\left(\tfrac{px}{y}\right)$$
   is an analytic function.
4. If $f(z) = u + iv = \cosh z$ is an analytic function, show that the curves $u =$ const. and $v =$ const. are orthogonal.
5. Find a, b if
   $$f(z) = (x^2 - 2xy + ay^2) + i\big(bx^2 - y^2 + 2xy\big)$$
   is an analytic function, given a and b are real, and also find $f(z)$ in terms of z.
6. If u is a harmonic function, show that u^2 is not a harmonic function unless u is a constant.
7. Find whether $u = y + e^x \cos y$ can be a real part of an analytic function.
8. If $u = x^2 - y^2$ and $v = \dfrac{-2y}{x^2 + y^2}$, prove that u and v are harmonic functions, but they are not harmonic conjugates to each other.
9. If $f(z) = u(r, \theta) + iv(r, \theta)$ is an analytic function, prove that
   $$r^2\frac{\partial^2 u}{\partial r^2} + r\frac{\partial u}{\partial r} + \frac{\partial^2 u}{\partial \theta^2} = 0$$
   (This equation is called Laplace's equation in polar form.)

"Mastery comes through practice"

3 Mappings or Transformations

Conformal Mapping and the Transformations $w = e^z$, $w = \sin z$, $w = \sinh z$

3.1 Mapping or Transformation

Under the mapping (or transformation) $w = f(z)$, every point (x, y) of the z-plane is mapped into some point (u, v) of the w-plane. Correspondingly, a curve C in the z-plane is mapped into a curve C' in the w-plane.

Definition 1: Conformal Mapping

A mapping $w = f(z)$ is said to be a **conformal mapping** if it preserves the angle between two curves *both in magnitude and in sense.*

Definition 2: Isogonal Mapping

If a mapping $w = f(z)$ preserves the angle between two curves *only in magnitude but not in sense*, then it is called an **isogonal mapping**.

Note

A mapping performed by an analytic function $f(z)$ is conformal at every point where $f'(z) \neq 0$. Points where $f'(z) = 0$ are called *critical points*; conformality fails there.

3.2 Some Standard Transformations

The transformation $w = e^z$

The function $w = e^z$ is analytic everywhere, and $\dfrac{dw}{dz} = e^z \neq 0$, so the mapping is conformal at every point. Writing w in polar form $w = Re^{i\varphi}$ and $z = x + iy$,

$$\begin{aligned} w = e^z = e^{x+iy} &= e^x e^{iy} \\ \Rightarrow \quad Re^{i\varphi} &= e^x e^{iy} \\ \Rightarrow \quad R = e^x \quad &\text{and} \quad \varphi = y. \end{aligned}$$

Case (i): Lines parallel to the y-axis, $x = k_1$.

Then

$$R = e^{k_1} \;\Rightarrow\; \sqrt{u^2 + v^2} = c_1 \;\Rightarrow\; u^2 + v^2 = c_1^2, \qquad c_1 = e^{k_1}.$$

This is a circle with centre $(0, 0)$ and radius $c_1 = e^{k_1}$.

Result

A line parallel to the y-axis in the z-plane is mapped into a circle in the w-plane. Hence lines parallel to the y-axis are mapped into **concentric circles** centred at the origin of the w-plane, such that

- $R = 1$ when $k_1 = 0$ (the unit circle),

- $R > 1$ when $k_1 > 0$,
- $0 < R < 1$ when $k_1 < 0$.

Case (ii): Lines parallel to the x-axis, $y = m_1$.

Then

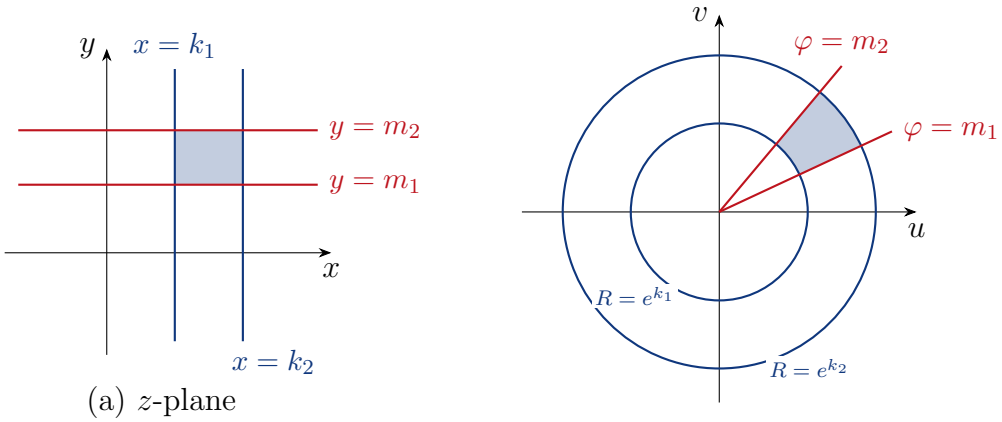
$$\varphi = m_1 \;\Rightarrow\; \tan^{-1}\left(\frac{v}{u}\right) = m_1 \;\Rightarrow\; \frac{v}{u} = \tan m_1 \;\Rightarrow\; v = c\,u, \qquad c = \tan m_1,$$

which is a straight line through the origin of the w-plane. Since $R > 0$ and φ is fixed, the image is actually the *half-line* (ray) from the origin making the angle m_1 with the positive u-axis.

Result

A line parallel to the x-axis in the z-plane is mapped into a radial line (ray) in the w-plane. Hence lines parallel to the x-axis are mapped into **radial lines** through the origin of the w-plane.

Hence, under $w = e^z$, the rectangular region bounded by the lines $x = k_1$, $x = k_2$ and $y = m_1$, $y = m_2$ in the z-plane (Figure 1(a)) is mapped onto the region bounded by the circles $R = e^{k_1}$, $R = e^{k_2}$ and the rays $\varphi = m_1$, $\varphi = m_2$ in the w-plane (Figure 1(b)).



(a) z-plane (b) w-plane

Figure 1: Region mapped under $w = e^z$.

The transformation $w = \sin z$

Given the transformation $w = \sin z$,

$$\begin{aligned} u + iv &= \sin(x + iy) \\ &= \sin x \cos(iy) + \cos x \sin(iy) \\ &= \sin x \cosh y + i \cos x \sinh y \\ \Rightarrow \quad u &= \sin x \cosh y, \qquad v = \cos x \sinh y. \end{aligned} \tag{1}$$

From (1),

$$\frac{u}{\cosh y} = \sin x \qquad \text{and} \qquad \frac{v}{\sinh y} = \cos x. \tag{2}$$

Using $\sin^2 x + \cos^2 x = 1$, we get

$$\frac{u^2}{\cosh^2 y} + \frac{v^2}{\sinh^2 y} = 1. \tag{3}$$

Case (i): Lines parallel to the x-axis, $y = k$ $(k \neq 0)$.

From (3),

$$\frac{u^2}{\cosh^2 k} + \frac{v^2}{\sinh^2 k} = 1 \quad \Longrightarrow \quad \frac{u^2}{a^2} + \frac{v^2}{b^2} = 1, \qquad a = \cosh k,\ b = \sinh k.$$

This represents an ellipse in the w-plane. Since $a^2 - b^2 = \cosh^2 k - \sinh^2 k = 1$, its foci are at $(\pm 1, 0)$ for every k.

Result

A line parallel to the x-axis in the z-plane is mapped into an ellipse in the w-plane. Hence lines parallel to the x-axis are mapped into **confocal ellipses** with foci $(\pm 1, 0)$.

Case (ii): Lines parallel to the y-axis, $x = m$ $(m \neq n\pi/2)$.

From (1),

$$\cosh y = \frac{u}{\sin x}, \qquad \sinh y = \frac{v}{\cos x}.$$

Using $\cosh^2 y - \sinh^2 y = 1$, we obtain

$$\frac{u^2}{\sin^2 x} - \frac{v^2}{\cos^2 x} = 1.$$

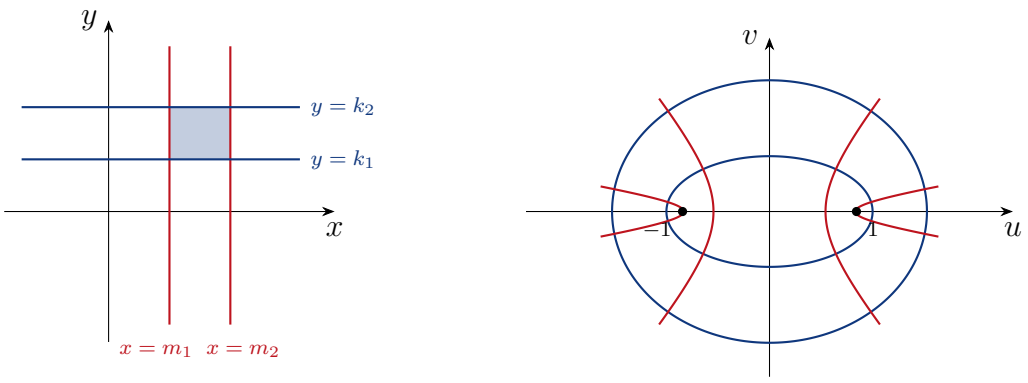
So, for $x = m$ (a constant),

$$\frac{u^2}{\sin^2 m} - \frac{v^2}{\cos^2 m} = 1 \quad \Longrightarrow \quad \frac{u^2}{p^2} - \frac{v^2}{q^2} = 1, \qquad p = \sin m,\ q = \cos m.$$

This represents a hyperbola in the w-plane. Since $p^2 + q^2 = \sin^2 m + \cos^2 m = 1$, its foci are also at $(\pm 1, 0)$.

Result

A line parallel to the y-axis is mapped into a hyperbola in the w-plane. Hence lines parallel to the y-axis are mapped into **confocal hyperbolas** with foci $(\pm 1, 0)$, the same foci as the ellipses of Case (i).



(a) z-plane (b) w-plane

Figure 2: Under $w = \sin z$: horizontal lines $\to$ confocal ellipses, vertical lines $\to$ confocal hyperbolas (foci ± 1).

3.3 Worked Example: The transformation $w = \sinh z$

Question 1: Discuss the transformation $w = \sinh z$

Discuss the transformation $w = \sinh z$ and find the images of lines parallel to the coordinate axes.

Solution. The function $w = \sinh z$ is analytic everywhere and $\dfrac{dw}{dz} = \cosh z$, which vanishes only at $z = i\left(\frac{\pi}{2} + n\pi\right)$, $n \in \mathbb{Z}$. Hence the mapping is conformal at all other points. Now

$$\begin{aligned} u + iv &= \sinh(x + iy) \\ &= \sinh x \cosh(iy) + \cosh x \sinh(iy) \\ &= \sinh x \cos y + i \cosh x \sin y \qquad (\because \cosh(iy) = \cos y,\ \sinh(iy) = i \sin y) \\ \Rightarrow \quad u &= \sinh x \cos y, \qquad v = \cosh x \sin y. \end{aligned} \tag{4}$$

Case (i): Lines parallel to the y-axis, $x = c$ $(c \neq 0)$.

From (4),

$$\cos y = \frac{u}{\sinh c}, \qquad \sin y = \frac{v}{\cosh c}.$$

Using $\cos^2 y + \sin^2 y = 1$,

$$\frac{u^2}{\sinh^2 c} + \frac{v^2}{\cosh^2 c} = 1 \quad \Longrightarrow \quad \frac{u^2}{b^2} + \frac{v^2}{a^2} = 1, \qquad a = \cosh c,\ b = \sinh c.$$

Since $a > b$, this is an ellipse in the w-plane with its *major axis along the v-axis*. Moreover $a^2 - b^2 = \cosh^2 c - \sinh^2 c = 1$, so the foci are at $(0, \pm 1)$ for every c.

Result

Lines parallel to the y-axis in the z-plane are mapped into **confocal ellipses** in the w-plane with foci $(0, \pm 1)$ (major axis along the v-axis).

Case (ii): Lines parallel to the x-axis, $y = k$ $(k \neq n\pi/2)$.

From (4),

$$\sinh x = \frac{u}{\cos k}, \qquad \cosh x = \frac{v}{\sin k}.$$

Using $\cosh^2 x - \sinh^2 x = 1$,

$$\frac{v^2}{\sin^2 k} - \frac{u^2}{\cos^2 k} = 1 \quad \Longrightarrow \quad \frac{v^2}{p^2} - \frac{u^2}{q^2} = 1, \qquad p = \sin k,\ q = \cos k.$$

This is a hyperbola in the w-plane with its *transverse axis along the v-axis*. Since $p^2 + q^2 = \sin^2 k + \cos^2 k = 1$, its foci are at $(0, \pm 1)$.

Result

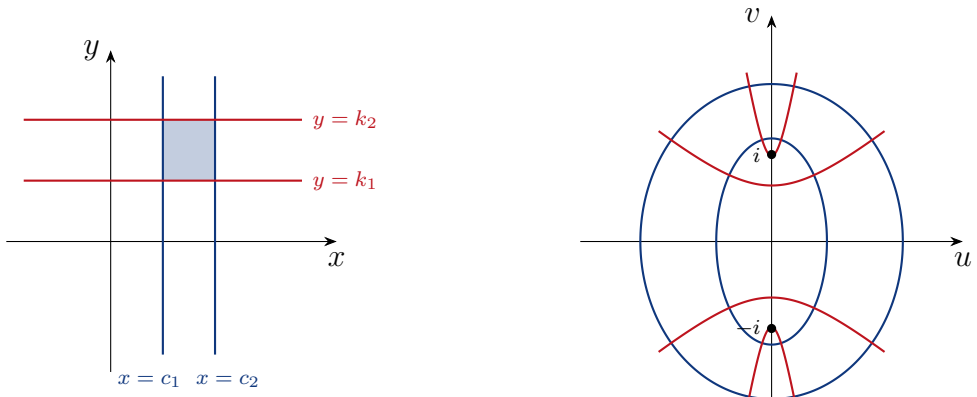
Lines parallel to the x-axis in the z-plane are mapped into **confocal hyperbolas** in the w-plane with foci $(0, \pm 1)$, the same foci as the ellipses of Case (i). The two families cut each other at right angles, as expected of a conformal map.

Special (degenerate) cases.

- $x = 0$ (the y-axis): $u = 0,\ v = \sin y$. The image is the segment $-1 \le v \le 1$ of the v-axis (traversed repeatedly).
- $y = 0$ (the x-axis): $u = \sinh x,\ v = 0$. The image is the *entire* u-axis.
- $y = \pi/2$: $u = 0,\ v = \cosh x \ge 1$. The image is the part $v \ge 1$ of the v-axis (a degenerate hyperbola).

Remark

Since $\sinh z = -i \sin(iz)$, the map $w = \sinh z$ is obtained from $w = \sin z$ by a rotation of the z-plane through $90°$ ($z \mapsto iz$) followed by a rotation of the w-plane through $-90°$ (multiplication by $-i$). This is why the pattern of Figure 2 reappears here with the roles of the axes interchanged: the foci move from $(\pm 1, 0)$ to $(0, \pm 1)$.



(a) z-plane

(b) w-plane

Figure 3: Under $w = \sinh z$: vertical lines $\to$ confocal ellipses, horizontal lines $\to$ confocal hyperbolas (foci $\pm i$).

3.4 Exercise

Exercise 1: $w = \cos z$

Discuss the transformation $w = \cos z$. Show that lines parallel to the x-axis are mapped into confocal ellipses and lines parallel to the y-axis into confocal hyperbolas in the w-plane, and find their common foci.

Exercise 2: $w = \cosh z$

Discuss the transformation $w = \cosh z$. Find the images of the lines $x =$ constant and $y =$ constant in the w-plane, and identify the degenerate cases $x = 0$ and $y = 0$.

4 The Transformation $w = 1/z$ and Bilinear Transformations

Images of Circles, Lines and Regions; Cross-Ratio and Fixed Points

4.1 Conformal Mapping: The Transformation $w = \frac{1}{z}$

Images of Circles, Lines and Regions

The Transformation $w = \frac{1}{z}$

Basic facts

Clearly $w = \dfrac{1}{z}$ is an analytic function (for $z \neq 0$). Writing $z = x + iy$ and $w = u + iv$,

$$u + iv = \frac{1}{x+iy} = \frac{x-iy}{x^2+y^2} \quad \Longrightarrow \quad u = \frac{x}{x^2+y^2}, \qquad v = \frac{-y}{x^2+y^2}.$$

Conversely, since $z = \dfrac{1}{w} = \dfrac{1}{u+iv} = \dfrac{u-iv}{u^2+v^2}$,

$$x = \frac{u}{u^2+v^2}, \qquad y = \frac{-v}{u^2+v^2}.$$

Note

Under the transformation $w = \dfrac{1}{z}$, circles and straight lines are mapped into either circles or straight lines. We verify this through some problems.

Recall. In complex form, the equation of a circle can be written as $|z - a| = r$; it is a circle with centre a and radius r. Also $|x + iy|^2 = x^2 + y^2$.

Solved Problems

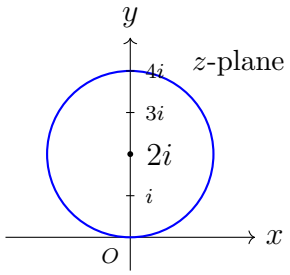
Problem 1

Find the image of the circle $|z - 2i| = 2$ under the transformation $w = \dfrac{1}{z}$.

Solution. The given circle is

$$|z - 2i| = 2. \tag{1}$$

Hence (1) is a circle with centre at $2i$ and radius 2 units (it passes through the origin).



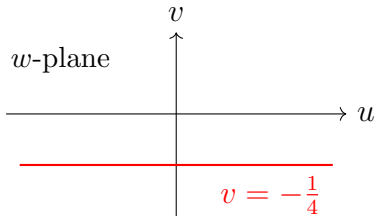
The transformation is $w = \dfrac{1}{z}$, i.e. $z = \dfrac{1}{w}$. Hence the given circle becomes

$$\left|\frac{1}{w} - 2i\right| = 2 \implies |1 - 2iw| = 2|w|$$
$$\implies |1 - 2i(u+iv)| = 2|u+iv| \implies |(1+2v) - 2iu| = 2|u+iv|.$$

Squaring both sides, we get

$$(1+2v)^2 + 4u^2 = 4(u^2+v^2)$$
$$\implies 1 + 4v^2 + 4v + 4u^2 = 4u^2 + 4v^2$$
$$\implies 4v + 1 = 0 \implies v = -\tfrac{1}{4}.$$

The image is the straight line $v = -\dfrac{1}{4}$, parallel to the u-axis in the w-plane.



Problem 2

Find the image of the circle $|z - 3i| = 2$ under $w = \dfrac{1}{z}$.

Solution. Given $w = \dfrac{1}{z}$ and $|z - 3i| = 2$. Then

$$\left|\frac{1}{w} - 3i\right| = 2 \implies |1 - 3iw| = 2|w| \implies |1 - 3i(u+iv)| = 2|u+iv|$$
$$\implies |(1+3v) - 3iu| = 2|u+iv|.$$

Squaring both sides, we get

$$(1+3v)^2 + (3u)^2 = 4(u^2+v^2)$$
$$1 + 9v^2 + 6v + 9u^2 = 4u^2 + 4v^2$$
$$5u^2 + 5v^2 + 6v + 1 = 0, \tag{2}$$

which is the equation of a circle. To get its centre and radius, (2) can be written as

$$u^2+v^2+\frac{6}{5}v+\frac{1}{5}=0 \implies (u-0)^2+\left(v^2-2bv\right)=-\frac{1}{5}, \quad \text{where } -2b=\frac{6}{5}\Rightarrow b=-\frac{3}{5}.$$

Completing the square,

$$(u-0)^2+\left[\left(v+\tfrac{3}{5}\right)^2-\tfrac{9}{25}\right]=-\frac{1}{5}$$
$$\implies (u-0)^2+\left(v-\left(-\tfrac{3}{5}\right)\right)^2=-\frac{1}{5}+\frac{9}{25}=\frac{4}{25}=\left(\frac{2}{5}\right)^2,$$

which is of the form $(u-a)^2+(v-b)^2=r^2$.

The image is a circle with centre $\left(0,-\frac{3}{5}\right)$ and radius $\frac{2}{5}$.

Problem 3

Find the images of the following regions under $w=\frac{1}{z}$:

(a) the half-plane $x>c,\ \ c>0$;

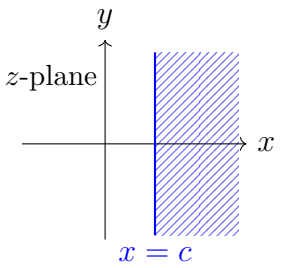
(b) the infinite strip $\frac{1}{4}\le y\le\frac{1}{2}$;

(c) the half-plane $x>c,\ \ c<0$.

Solution. Given $w=\frac{1}{z}$. Under this transformation we have

$$z=\frac{1}{w}=\frac{1}{u+iv}=\frac{u-iv}{u^2+v^2} \implies x=\frac{u}{u^2+v^2} \quad \text{and} \quad y=\frac{-v}{u^2+v^2}.$$

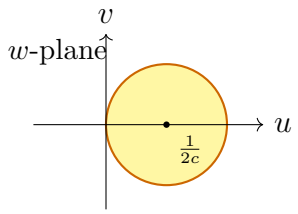
(a) The half-plane $x>c$, $c>0$



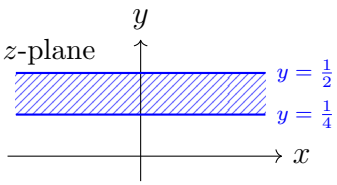
$$\begin{aligned}
x>c &\implies \frac{u}{u^2+v^2}>c\\
&\implies u>c\,(u^2+v^2)\\
&\implies u^2+v^2<\frac{u}{c}\\
&\implies u^2-\frac{u}{c}+v^2<0\\
&\implies \left(u-\frac{1}{2c}\right)^2-\frac{1}{4c^2}+v^2<0\\
&\implies \left(u-\frac{1}{2c}\right)^2+(v-0)^2<\left(\frac{1}{2c}\right)^2.
\end{aligned}$$

Here $u^2-\frac{u}{c}=A^2-2AB$ with $A=u$ and $2AB=\frac{u}{c}\Rightarrow B=\frac{1}{2c}$.
So $u^2-\frac{u}{c}=(A-B)^2-B^2$.

The image is the region *inside* the circle with centre $\left(\frac{1}{2c}, 0\right)$ and radius $\frac{1}{2c}$.



(b) The infinite strip $\frac{1}{4} \leq y \leq \frac{1}{2}$



Now $y \geq \frac{1}{4}$:

$$\begin{aligned} &\frac{-v}{u^2+v^2} \geq \frac{1}{4} \\ \Rightarrow\ &-4v \geq u^2+v^2 \\ \Rightarrow\ &u^2+v^2+4v \leq 0 \\ \Rightarrow\ &u^2+(v+2)^2-4 \leq 0 \\ \Rightarrow\ &(u-0)^2+\big(v-(-2)\big)^2 \leq 2^2, \end{aligned}$$

a region *inside* the circle with centre $(0,-2)$ and radius 2.

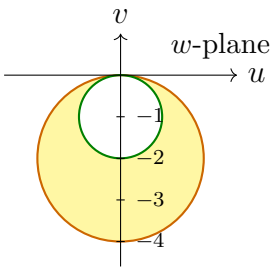
Now $y \leq \frac{1}{2}$:

$$\begin{aligned} &\frac{-v}{u^2+v^2} \leq \frac{1}{2} \\ \Rightarrow\ &-2v \leq u^2+v^2 \\ \Rightarrow\ &u^2+v^2+2v \geq 0 \\ \Rightarrow\ &u^2+(v+1)^2-1 \geq 0 \\ \Rightarrow\ &(u-0)^2+\big(v-(-1)\big)^2 \geq 1, \end{aligned}$$

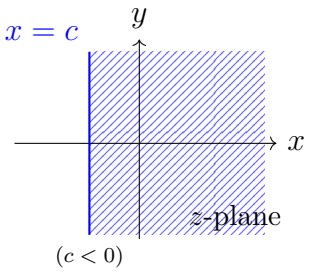
a region *outside* the circle with centre $(0,-1)$ and radius 1.

Combining these two regions, we get:

The image of the strip is the crescent-shaped region lying inside the circle $u^2+(v+2)^2=4$ and outside the circle $u^2+(v+1)^2=1$.



(c) The half-plane $x > c$, $c < 0$



Under $w = \dfrac{1}{z}$,

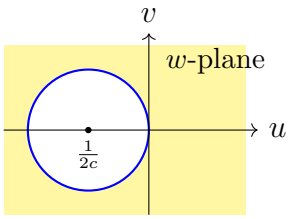
$$x > c \implies \frac{u}{u^2+v^2} > c.$$

Since $c < 0$, put $c = -k$ with $k > 0$.

$$\begin{aligned}
&\frac{u}{u^2+v^2} > -k \\
\Rightarrow\ &\frac{u}{k} > -(u^2+v^2) \\
\Rightarrow\ &-\frac{u}{k} < u^2+v^2 \\
\Rightarrow\ &u^2+v^2+\frac{u}{k} > 0 \\
\Rightarrow\ &\left(u^2+\frac{u}{k}\right)+v^2 > 0 \\
\Rightarrow\ &\left(u+\frac{1}{2k}\right)^2-\frac{1}{4k^2}+v^2 > 0 \\
\Rightarrow\ &\left(u-\left(-\tfrac{1}{2k}\right)\right)^2+(v-0)^2 > \left(\frac{1}{2k}\right)^2 \\
\Rightarrow\ &\left(u-\frac{1}{2c}\right)^2+(v-0)^2 > \left(\frac{1}{-2c}\right)^2 \quad (\because k=-c).
\end{aligned}$$

Here $u^2+\dfrac{u}{k} = A^2+2AB$ with $A = u$ and $2AB = \dfrac{u}{k} \Rightarrow B = \dfrac{1}{2k}$.
So $u^2+\dfrac{u}{k} = (A+B)^2 - B^2$.

The image is the region *outside* the circle with centre $\left(\dfrac{1}{2c}, 0\right)$ and radius $\dfrac{1}{2|c|} = -\dfrac{1}{2c}$ (the radius is positive since $c < 0$).



Exercise

Problem 4

Find the image of the circle $|z| = 3$ under $w = \dfrac{1}{z}$.

4.2 Bilinear Transformation

Conformality, Cross-Ratio, Mapping of Regions and Fixed Points

Bilinear Transformation (BT)

Definition

A **bilinear transformation** (also called a **Möbius transformation** or a **linear fractional transformation**) is a mapping of the complex z-plane into the complex w-plane of the form

$$w = f(z) = \frac{az+b}{cz+d},$$

where a, b, c, d are complex constants satisfying

$$ad - bc \neq 0.$$

Why $ad - bc \neq 0$? If $ad - bc = 0$ and $c \neq 0$, then $b = ad/c$ and

$$w = \frac{az + ad/c}{cz+d} = \frac{a}{c} \cdot \frac{cz+d}{cz+d} = \frac{a}{c},$$

a constant. (If $c = 0$, then $ad = 0$, so either $d = 0$ and the map is undefined, or $a = 0$ and $w = b/d$ is constant.) Thus every point would be mapped to one point: the transformation is *degenerate*.

Inverse. Solving $w(cz + d) = az + b$ for z gives

$$z = \frac{dw - b}{-cw + a},$$

which is again bilinear, since $da - (-b)(-c) = ad - bc \neq 0$.

Condition for Conformality

Differentiate $w = \dfrac{az+b}{cz+d}$ by the quotient rule:

$$\frac{dw}{dz} = \frac{a(cz+d) - c(az+b)}{(cz+d)^2} = \frac{acz + ad - acz - bc}{(cz+d)^2} = \frac{ad-bc}{(cz+d)^2}.$$

Result

Since $ad - bc \neq 0$, $\dfrac{dw}{dz} \neq 0$ at every point where it exists. Hence a bilinear transformation is **conformal at every point** $z \neq -d/c$. At $z = -d/c$ (when $c \neq 0$) the function is not analytic; this point is mapped to $w = \infty$.

Note

The derivative of a (non-degenerate) BT *never* vanishes. The only point of the finite plane where conformality fails is the pole $z = -d/c$. If $ad - bc = 0$, then $dw/dz \equiv 0$, the map is constant, and there is no question of conformality.

Cross-Ratio

Definition

The **cross-ratio** of four distinct points z_1, z_2, z_3, z_4 is

$$(z_1, z_2, z_3, z_4) = \frac{(z_1 - z_2)(z_3 - z_4)}{(z_1 - z_4)(z_3 - z_2)}.$$

Similarly, for four points w_1, w_2, w_3, w_4 of the w-plane,

$$(w_1, w_2, w_3, w_4) = \frac{(w_1 - w_2)(w_3 - w_4)}{(w_1 - w_4)(w_3 - w_2)}.$$

Theorem (Invariance of the cross-ratio)

If a bilinear transformation maps z_1, z_2, z_3, z_4 to w_1, w_2, w_3, w_4 respectively, then

$$(w_1, w_2, w_3, w_4) = (z_1, z_2, z_3, z_4).$$

Proof. Let $w_k = \dfrac{az_k + b}{cz_k + d}$. Then

$$w_i - w_j = \frac{(az_i + b)(cz_j + d) - (az_j + b)(cz_i + d)}{(cz_i + d)(cz_j + d)} = \frac{(ad - bc)(z_i - z_j)}{(cz_i + d)(cz_j + d)}.$$

Substituting this in (w_1, w_2, w_3, w_4), the factor $(ad - bc)^2$ cancels, and each of the factors $(cz_1 + d), (cz_2 + d), (cz_3 + d), (cz_4 + d)$ occurs exactly once in the numerator and once in the denominator, so they cancel too. What remains is $\dfrac{(z_1 - z_2)(z_3 - z_4)}{(z_1 - z_4)(z_3 - z_2)} = (z_1, z_2, z_3, z_4)$. ■

Finding a BT using the cross-ratio

A BT is determined uniquely by the images of three distinct points. Suppose the three given points z_2, z_3, z_4 are to be mapped to w_2, w_3, w_4, and let z_1 be any other point with image w_1. By invariance, $(w_1, w_2, w_3, w_4) = (z_1, z_2, z_3, z_4)$.

Working formula

$$\frac{(w_1 - w_2)(w_3 - w_4)}{(w_1 - w_4)(w_3 - w_2)} = \frac{(z_1 - z_2)(z_3 - z_4)}{(z_1 - z_4)(z_3 - z_2)}$$

Put $w_1 = w$ and $z_1 = z$ (the variable point and its image):

$$\frac{(w - w_2)(w_3 - w_4)}{(w - w_4)(w_3 - w_2)} = \frac{(z - z_2)(z_3 - z_4)}{(z - z_4)(z_3 - z_2)}$$

Solving this equation for w gives the required transformation $w = f(z)$.

Note

Points at infinity. If one of the given points is ∞, the two factors containing it are replaced by their limiting ratio, which is 1. For example, if $z_4 = \infty$,

$$\frac{z_3 - z_4}{z - z_4} \longrightarrow 1 \quad \text{as } z_4 \to \infty, \qquad \text{so} \qquad (z, z_2, z_3, \infty) = \frac{z - z_2}{z_3 - z_2}.$$

Similarly, if $w_4 = \infty$, then $(w, w_2, w_3, \infty) = \dfrac{w - w_2}{w_3 - w_2}$. Expressions such as $2 - \infty$ must never be written.

Problems: Finding the Bilinear Transformation

Problem 1

Find the BT which maps $z_2 = 1,\ z_3 = 2,\ z_4 = 3$ onto $w_2 = 4,\ w_3 = 5,\ w_4 = 6$.

Solution. The working formula is

$$\frac{(w_1 - w_2)(w_3 - w_4)}{(w_1 - w_4)(w_3 - w_2)} = \frac{(z_1 - z_2)(z_3 - z_4)}{(z_1 - z_4)(z_3 - z_2)}.$$

Put $w_1 = w$ and $z_1 = z$:

$$\frac{(w - w_2)(w_3 - w_4)}{(w - w_4)(w_3 - w_2)} = \frac{(z - z_2)(z_3 - z_4)}{(z - z_4)(z_3 - z_2)}.$$

Substituting the given values,

$$\frac{(w-4)(5-6)}{(w-6)(5-4)} = \frac{(z-1)(2-3)}{(z-3)(2-1)} \implies \frac{-(w-4)}{w-6} = \frac{-(z-1)}{z-3} \implies \frac{w-4}{w-6} = \frac{z-1}{z-3}.$$

Cross-multiplying,

$$\begin{aligned}(w-4)(z-3) &= (w-6)(z-1)\\ wz - 3w - 4z + 12 &= wz - w - 6z + 6\\ -3w + w &= -6z + 4z + 6 - 12\\ -2w &= -2z - 6.\end{aligned}$$

$$\boxed{w = z + 3}$$

Check: $1 \mapsto 4,\ 2 \mapsto 5,\ 3 \mapsto 6$. ✓

Problem 2

Find the BT which maps $z_2 = 2,\ z_3 = 3,\ z_4 = 5$ onto $w_2 = 1,\ w_3 = 2,\ w_4 = 3$.

Solution. The working formula is

$$\frac{(w_1 - w_2)(w_3 - w_4)}{(w_1 - w_4)(w_3 - w_2)} = \frac{(z_1 - z_2)(z_3 - z_4)}{(z_1 - z_4)(z_3 - z_2)}.$$

Put $w_1 = w$ and $z_1 = z$:

$$\frac{(w-w_2)(w_3-w_4)}{(w-w_4)(w_3-w_2)} = \frac{(z-z_2)(z_3-z_4)}{(z-z_4)(z_3-z_2)}.$$

Substituting the given values,

$$\frac{(w-1)(2-3)}{(w-3)(2-1)} = \frac{(z-2)(3-5)}{(z-5)(3-2)} \quad\Longrightarrow\quad \frac{-(w-1)}{w-3} = \frac{-2(z-2)}{z-5}$$

$$\Longrightarrow\quad \frac{w-1}{w-3} = \frac{2(z-2)}{z-5}.$$

Cross-multiplying,

$$\begin{aligned}(w-1)(z-5) &= 2(z-2)(w-3)\\ wz-5w-z+5 &= 2wz-6z-4w+12\\ 0 &= wz+w-5z+7\\ w(z+1) &= 5z-7.\end{aligned}$$

$$\boxed{w = \frac{5z-7}{z+1}}$$

Check: $w(2) = \frac{3}{3} = 1,\ w(3) = \frac{8}{4} = 2,\ w(5) = \frac{18}{6} = 3.$ ✓

Problem 3

Find the BT which maps $z_2 = 1,\ z_3 = 2,\ z_4 = \infty$ onto $w_2 = 0,\ w_3 = 1,\ w_4 = \infty$.

Solution. The working formula is

$$\frac{(w_1-w_2)(w_3-w_4)}{(w_1-w_4)(w_3-w_2)} = \frac{(z_1-z_2)(z_3-z_4)}{(z_1-z_4)(z_3-z_2)}.$$

Put $w_1 = w$ and $z_1 = z$:

$$\frac{(w-w_2)(w_3-w_4)}{(w-w_4)(w_3-w_2)} = \frac{(z-z_2)(z_3-z_4)}{(z-z_4)(z_3-z_2)}.$$

Rewrite it as

$$\frac{w-w_2}{w_3-w_2}\cdot\frac{w_3-w_4}{w-w_4} = \frac{z-z_2}{z_3-z_2}\cdot\frac{z_3-z_4}{z-z_4}.$$

Here $z_4 = \infty$ and $w_4 = \infty$. Dividing numerator and denominator by z_4 (and by w_4),

$$\frac{z_3-z_4}{z-z_4} = \frac{z_3/z_4-1}{z/z_4-1} \longrightarrow \frac{0-1}{0-1} = 1, \qquad \frac{w_3-w_4}{w-w_4} \longrightarrow 1.$$

Hence the formula reduces to

$$\frac{w-w_2}{w_3-w_2} = \frac{z-z_2}{z_3-z_2} \quad\Longrightarrow\quad \frac{w-0}{1-0} = \frac{z-1}{2-1}.$$

$$\boxed{w = z-1}$$

Check: $1 \mapsto 0,\ 2 \mapsto 1,\ \infty \mapsto \infty.$ ✓

Problem 4

Find the BT which maps $z_2 = 0,\ z_3 = 1,\ z_4 = \infty$ onto $w_2 = -1,\ w_3 = 1,\ w_4 = \infty$.

Solution. The working formula is

$$\frac{(w_1 - w_2)(w_3 - w_4)}{(w_1 - w_4)(w_3 - w_2)} = \frac{(z_1 - z_2)(z_3 - z_4)}{(z_1 - z_4)(z_3 - z_2)}.$$

Put $w_1 = w$ and $z_1 = z$:

$$\frac{(w - w_2)(w_3 - w_4)}{(w - w_4)(w_3 - w_2)} = \frac{(z - z_2)(z_3 - z_4)}{(z - z_4)(z_3 - z_2)}.$$

Since $z_4 = w_4 = \infty$, the factors $\dfrac{z_3 - z_4}{z - z_4}$ and $\dfrac{w_3 - w_4}{w - w_4}$ tend to 1 (as in Problem 3), and the formula reduces to

$$\frac{w - w_2}{w_3 - w_2} = \frac{z - z_2}{z_3 - z_2} \quad\Longrightarrow\quad \frac{w - (-1)}{1 - (-1)} = \frac{z - 0}{1 - 0} \quad\Longrightarrow\quad \frac{w + 1}{2} = z.$$

$$w = 2z - 1$$

Check: $0 \mapsto -1,\ 1 \mapsto 1,\ \infty \mapsto \infty$. ✓

Problem 5

Find the BT which maps $z_2 = 0,\ z_3 = 1,\ z_4 = -1$ onto $w_2 = 1,\ w_3 = -1,\ w_4 = 0$.

Solution. The working formula is

$$\frac{(w_1 - w_2)(w_3 - w_4)}{(w_1 - w_4)(w_3 - w_2)} = \frac{(z_1 - z_2)(z_3 - z_4)}{(z_1 - z_4)(z_3 - z_2)}.$$

Put $w_1 = w$ and $z_1 = z$:

$$\frac{(w - w_2)(w_3 - w_4)}{(w - w_4)(w_3 - w_2)} = \frac{(z - z_2)(z_3 - z_4)}{(z - z_4)(z_3 - z_2)}.$$

Substituting the given values,

$$\frac{(w - 1)(-1 - 0)}{(w - 0)(-1 - 1)} = \frac{(z - 0)(1 + 1)}{(z + 1)(1 - 0)} \quad\Longrightarrow\quad \frac{w - 1}{2w} = \frac{2z}{z + 1}.$$

Cross-multiplying,

$$\begin{aligned}(w - 1)(z + 1) &= 4zw\\ wz + w - z - 1 &= 4wz\\ w - 3wz &= z + 1\\ w(1 - 3z) &= z + 1.\end{aligned}$$

$$w = \frac{z + 1}{1 - 3z}$$

Check: $w(0) = 1,\ w(1) = \frac{2}{-2} = -1,\ w(-1) = 0$. ✓

Mapping of Regions

Throughout, $w = u + iv$.

Problem 6

Show that the BT $w = \dfrac{1+z}{1-z}$ maps the region $|z| \le 1$ onto the region $\mathrm{Re}(w) \ge 0$ of the w-plane.

Solution. Inverse. From $w(1-z) = 1+z$ we get $w - wz = 1 + z$, i.e. $w - 1 = z(w+1)$. Hence

$$z = \frac{w-1}{w+1} \qquad (w \neq -1).$$

($w = -1$ corresponds to $z = \infty$, which is not in the region $|z| \le 1$.)

Given region. $|z| \le 1$ gives

$$\left|\frac{w-1}{w+1}\right| \le 1 \quad \Longrightarrow \quad |w-1| \le |w+1| \qquad (\text{as } |w+1| > 0).$$

Put $w = u + iv$:

$$|(u-1)+iv| \le |(u+1)+iv| \quad \Longrightarrow \quad \sqrt{(u-1)^2+v^2} \le \sqrt{(u+1)^2+v^2}.$$

Squaring both sides (both sides are non-negative, so the inequality is preserved):

$$\begin{aligned} (u-1)^2 + v^2 &\le (u+1)^2 + v^2 \\ u^2 - 2u + 1 + v^2 &\le u^2 + 2u + 1 + v^2 \\ -2u &\le 2u \\ -4u \le 0 \quad &\Longrightarrow \quad u \ge 0. \end{aligned}$$

$$|z| \le 1 \;\longmapsto\; \mathrm{Re}(w) = u \ge 0$$

In particular, the circle $|z| = 1$ is mapped onto the imaginary axis $u = 0$ (the point $z = 1$ goes to $w = \infty$), and the interior $|z| < 1$ onto the right half-plane $u > 0$.

Problem 7

Show that the transformation $w = \dfrac{z-i}{1-iz}$ maps the interior of the unit circle $|z| < 1$ onto the lower half of the w-plane.

Solution. Inverse. From $w(1-iz) = z - i$ we get $w - iwz = z - i$, i.e. $w + i = z(1 + iw)$. Hence

$$z = \frac{w+i}{1+iw} \qquad (w \neq i).$$

Given region. $|z| < 1$ gives

$$\left|\frac{w+i}{1+iw}\right| < 1 \quad \Longrightarrow \quad |w+i| < |1+iw|.$$

Put $w = u + iv$:

$$w + i = u + i(v+1), \qquad 1 + iw = 1 + i(u+iv) = (1-v) + iu.$$

Therefore

$$|u + i(v+1)| < |(1-v) + iu| \quad \Longrightarrow \quad \sqrt{u^2 + (v+1)^2} < \sqrt{(1-v)^2 + u^2}.$$

Squaring both sides (both sides are non-negative):

$$\begin{aligned} u^2 + (v+1)^2 &< (1-v)^2 + u^2 \\ u^2 + v^2 + 2v + 1 &< 1 - 2v + v^2 + u^2 \\ 2v &< -2v \\ 4v < 0 \quad &\Longrightarrow \quad v < 0. \end{aligned}$$

$$|z| < 1 \;\longmapsto\; \mathrm{Im}(w) = v < 0 \quad \text{(lower half-plane)}$$

Check: $z = 0 \mapsto w = -i$, which lies in the lower half-plane. ✓

Fixed Points (Invariant Points)

Definition

A point z is called a **fixed point** (or **invariant point**) of the transformation $w = f(z)$ if its image is the point itself, i.e. $f(z) = z$.

For $w = \dfrac{az+b}{cz+d}$, putting $w = z$ gives $z(cz+d) = az+b$, i.e.

$$cz^2 + (d-a)z - b = 0.$$

Result

A bilinear transformation (other than the identity $w = z$) has at most two fixed points, namely the roots of $cz^2 + (d-a)z - b = 0$. If $c = 0$, one of the fixed points is $z = \infty$.

Problem 8

Find the fixed points of the transformation $w = \dfrac{iz-4}{z-2i}$.

Solution. Replacing w by z:

$$\begin{aligned} z &= \frac{iz-4}{z-2i} \\ z^2 - 2iz &= iz - 4 \\ z^2 - 3iz + 4 &= 0. \end{aligned}$$

By the quadratic formula (with $a = 1,\ b = -3i,\ c = 4$),

$$z = \frac{3i \pm \sqrt{(-3i)^2 - 4(1)(4)}}{2} = \frac{3i \pm \sqrt{-9-16}}{2} = \frac{3i \pm \sqrt{-25}}{2} = \frac{3i \pm 5i}{2}.$$

Hence

$$z = \frac{3i+5i}{2} = \frac{8i}{2} = 4i \qquad \text{or} \qquad z = \frac{3i-5i}{2} = \frac{-2i}{2} = -i.$$

$$z = 4i \quad \text{and} \quad z = -i$$

Check: $(z-4i)(z+i) = z^2 + iz - 4iz - 4i^2 = z^2 - 3iz + 4$. ✓

Also $w(4i) = \dfrac{i(4i)-4}{4i-2i} = \dfrac{-8}{2i} = 4i$ and $w(-i) = \dfrac{i(-i)-4}{-i-2i} = \dfrac{-3}{-3i} = \dfrac{1}{i} = -i$. ✓

Problem 9

Find the fixed points of the transformation $w = \dfrac{z-1}{z+1}$.

Solution. Replacing w by z:

$$z = \frac{z-1}{z+1} \implies z^2 + z = z - 1 \implies z^2 + 1 = 0 \implies z^2 = -1.$$

$$z = \pm i$$

Problem 10

Find all linear fractional transformations whose fixed points are 1 and -1.

Solution. Let $w = \dfrac{az+b}{cz+d}$, $ad - bc \neq 0$.

When $z = 1$ is fixed:

$$1 = \frac{a+b}{c+d} \quad \Longrightarrow \quad a + b = c + d \qquad (1)$$

When $z = -1$ is fixed:

$$-1 = \frac{-a+b}{-c+d} \quad \Longrightarrow \quad c - d = -a + b \qquad (2)$$

Adding (1) and (2): $2b = 2c$, so $b = c$. Substituting in (1): $a + c = c + d$, so $a = d$.

$$w = \frac{az+b}{bz+a}, \qquad a^2 - b^2 \neq 0$$

(For $b = 0$ this is the identity $w = z$, which fixes every point; for $b \neq 0$ the only fixed points are ± 1.)

Problem 11

Find all linear fractional transformations whose fixed points are i and $-i$.

Solution. Let $w = \dfrac{az+b}{cz+d}$, $ad - bc \neq 0$.

Case (i): $z = i$ is fixed:

$$i = \frac{ai+b}{ci+d} \implies i(ci+d) = ai + b \implies -c + di = ai + b \qquad (1)$$

Case (ii): $z = -i$ is fixed:

$$-i = \frac{-ai+b}{-ci+d} \implies -i(-ci+d) = -ai + b \implies -c - di = -ai + b \qquad (2)$$

Since a, b, c, d are *complex*, we may not equate real and imaginary parts. Instead:

$$\begin{aligned} (1) - (2): &\quad 2di = 2ai \implies a = d, \\ (1) + (2): &\quad -2c = 2b \implies c = -b. \end{aligned}$$

$$w = \frac{az+b}{-bz+a}, \qquad a^2 + b^2 \neq 0$$

(For $b = 0$ this is the identity; for $b \neq 0$ the only fixed points are $\pm i$.)

5 Exercise Problems

Mapping $w = 1/z$ and Bilinear Transformations

Exercise 1

Find the bilinear transformation which maps the points

(i) $z = i$, -1, 1 of the z-plane into the points $w = 0$, 1, ∞ of the w-plane, respectively;

(ii) $z = 0$, $z = 1$ and $z = \infty$ into the points $w = i$, $w = 1$ and $w = -i$, respectively.

Exercise 2

Find the image of the circle $|z - 3i| = 3$ under the mapping $w = \dfrac{1}{z}$.

Exercise 3

Show that the function $w = \dfrac{4}{z}$ transforms the straight line $x = c$ in the z-plane into a circle in the w-plane.

Exercise 4

Show that the transformation $w = i\,\dfrac{1 - z}{1 + z}$ transforms the circle $|z| = 1$ onto the real axis of the w-plane, and the interior of the circle into the upper half of the w-plane.

Exercise 5

Show that $w = \dfrac{i - z}{i + z}$ maps

(i) the real axis of the z-plane onto the circle $|w| = 1$, and

(ii) the half-plane $y > 0$ onto the interior of the unit circle $|w| < 1$ in the w-plane.

Exercise 6

Find the fixed points of the transformations

$$\text{(a)}\quad w = \frac{3z - 4}{z - 1} \qquad \text{and} \qquad \text{(b)}\quad w = \frac{z - 1}{z + 1}.$$

6 Physical Applications of Complex Differentiation

Fluid Flow, Electrostatics, AC Circuit Analysis and Heat Transfer

Syllabus

Analytic functions – Cauchy–Riemann equations and other properties – Harmonic functions – Construction of an analytic function by Milne–Thomson method only – Conformal mappings – Mappings $\sin z$, e^z, $1/z$ only – Möbius transformation – Analysing AC circuits – Heat transfer problems.

6.1 Quick Review of the Tools

Definition 6.1 : *Analytic function*

A function $f(z) = u(x,y) + i\,v(x,y)$, $z = x+iy$, is *analytic* at z_0 if it is differentiable at every point of some neighbourhood of z_0. It is analytic in a domain D if it is analytic at every point of D. $u = \operatorname{Re} f$ is the *real part* and $v = \operatorname{Im} f$ is the *imaginary part*.

Result 6.1 : *Cauchy–Riemann (C–R) equations*

If $f = u + iv$ is analytic, then

$$\text{Cartesian:}\quad u_x = v_y, \qquad u_y = -v_x; \qquad\qquad \text{Polar:}\quad u_r = \tfrac{1}{r}v_\theta, \qquad v_r = -\tfrac{1}{r}u_\theta.$$

Conversely, if u, v, u_x, u_y, v_x, v_y are continuous and satisfy the C–R equations in D, then f is analytic in D, and

$$f'(z) = u_x + i\,v_x = u_x - i\,u_y = v_y + i\,v_x.$$

Result 6.2 : *Properties used in every application*

1. **Both parts are harmonic:** $\nabla^2 u = u_{xx} + u_{yy} = 0$ and $\nabla^2 v = 0$.
2. **Orthogonality:** the level curves $u(x,y) = c_1$ and $v(x,y) = c_2$ cut each other at right angles (wherever $f'(z) \neq 0$), because $\nabla u \cdot \nabla v = u_x v_x + u_y v_y = -u_x u_y + u_y u_x = 0$.
3. **Gradient in complex form:** $\overline{f'(z)} = u_x + i\,u_y$, i.e. the complex number $\overline{f'(z)}$ *is* the vector ∇u, and $|f'(z)| = |\nabla u| = |\nabla v|$.

Definition 6.2 : *Harmonic function and harmonic conjugate*

A real function $\phi(x,y)$ with continuous second partial derivatives satisfying $\phi_{xx} + \phi_{yy} = 0$ is *harmonic*. If $u + iv$ is analytic, v is called a *harmonic conjugate* of u.

Result 6.3 : *Milne–Thomson method*

(a) Real part u given: $f'(z) = u_x(z,0) - i\,u_y(z,0)$, $f(z) = \int \big[u_x(z,0) - i\,u_y(z,0)\big]dz + C$.

(b) Imaginary part v given: $f'(z) = v_y(z,0) + i\,v_x(z,0)$, $f(z) = \int \big[v_y(z,0) + i\,v_x(z,0)\big]dz + C$.

(Replace x by z and y by 0 in the partial derivatives, then integrate with respect to z.)

Note

The justification: with $x = \frac{z+\bar{z}}{2}$, $y = \frac{z-\bar{z}}{2i}$, the identity $f'(z) = u_x(x,y) - iu_y(x,y)$ holds for all z; putting $\bar{z} = z$ (so $x = z$, $y = 0$) gives formula (a).

6.2 Why the Real and Imaginary Parts have Physical Meaning

Many two-dimensional steady physical fields are governed by **Laplace's equation** $\phi_{xx}+\phi_{yy} = 0$. Since the real and imaginary parts of an analytic function are harmonic (Result 6.2), *every analytic function describes some physical field*, and every physical field of this type can be packed into one analytic function

$$\boxed{\Omega(z) = \phi(x,y) + i\,\psi(x,y)}$$

called the **complex potential**. The real part ϕ is the *potential* (a scalar that is "level" along certain curves); the imaginary part ψ is the *stream / flux function* whose level curves show the *paths along which the physical quantity actually travels.*

The central idea

- $\phi =$ constant : **equipotential curves** (isotherms, equipotential lines, ...)
- $\psi =$ constant : **flow / flux lines** (streamlines, lines of force, heat-flow lines)
- The two families are **orthogonal**: the physical quantity always flows perpendicular to the equipotentials, i.e. along the direction of steepest change of ϕ.
- $\Omega'(z)$ gives the **field vector**: $\overline{\Omega'(z)} = \phi_x + i\,\phi_y = \nabla\phi$.
- The difference $\psi_B - \psi_A$ measures the **amount flowing** between the flux lines through A and B.

Note

Sign conventions for fluid velocity vary between textbooks ($\vec{q} = \nabla\phi$ or $\vec{q} = -\nabla\phi$). These notes use $\vec{q} = \nabla\phi$ for fluids, and the physical sign $\vec{E} = -\nabla V$, $\vec{q} = -k\nabla T$ for electric field and heat flux. In AC circuits (Section 6.5) the real and imaginary parts have a *different* but equally concrete meaning.

	Fluid flow	Electrostatics	Heat conduction
Why Laplace?	incompressible, irrotational flow	charge-free region (Gauss's law)	steady state, no heat sources
$\phi = \operatorname{Re}\Omega$	velocity potential	electrostatic potential V	temperature T
$\psi = \operatorname{Im}\Omega$	stream function	flux function	heat-flow function
$\phi = c$	equipotential lines	equipotential lines	isotherms
$\psi = c$	streamlines	lines of force	heat-flow lines
Field vector	velocity $\vec{q} = \overline{\Omega'(z)}$	$\vec{E} = -\overline{\Omega'(z)}$	heat flux $\vec{q} = -k\,\overline{\Omega'(z)}$
Magnitude	speed $= \lvert\Omega'(z)\rvert$	$\lvert\vec{E}\rvert = \lvert\Omega'(z)\rvert$	$\lvert\vec{q}\rvert = k\lvert\Omega'(z)\rvert$

Table 1: Dictionary of meanings of the real and imaginary parts of the complex potential $\Omega = \phi + i\psi$.

6.3 Application I : Two-Dimensional Fluid Flow

Definition 6.3 : *Velocity potential and stream function*

For a steady, incompressible, irrotational two-dimensional flow with velocity $\vec{q} = (q_1, q_2)$ there exist functions ϕ and ψ with

$$q_1 = \phi_x = \psi_y, \qquad q_2 = \phi_y = -\psi_x.$$

ϕ is the *velocity potential*, ψ the *stream function*, and $\Omega = \phi + i\psi$ is analytic (these two relations are exactly the C–R equations).

Result 6.4 : *Reading the flow from $\Omega(z)$*

1. Complex velocity: $q_1 + i\,q_2 = \overline{\Omega'(z)}$; speed $\lvert\vec{q}\rvert = \lvert\Omega'(z)\rvert$.
2. Fluid particles move along the **streamlines** $\psi = c$.
3. **Stagnation points** (fluid at rest) are the zeros of $\Omega'(z)$.
4. Volume of fluid crossing any curve joining A to B (per unit depth, per unit time) $= \lvert\psi_B - \psi_A\rvert$.
5. Any streamline may be replaced by a rigid wall without disturbing the flow.

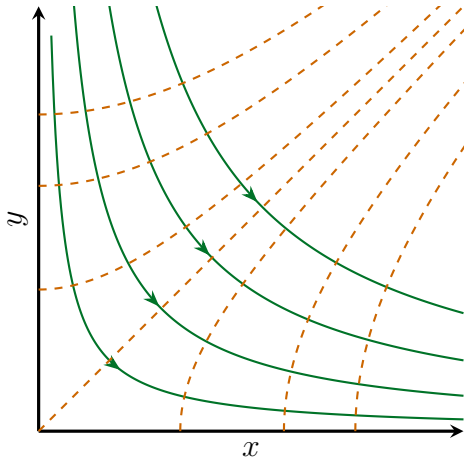
Problems on constructing the complex potential

Problem 6.1 : *Velocity potential given*

The velocity potential of a two-dimensional flow is $\phi = 3x^2y - y^3$. Verify that it is a possible potential, find the stream function ψ and the complex potential.

Flow	Complex potential $\Omega(z)$	ϕ	ψ
Uniform stream (speed U)	Uz	Ux	Uy
Flow in a 90° corner	z^2	$x^2 - y^2$	$2xy$
Source of strength m	$\frac{m}{2\pi}\operatorname{Log} z$	$\frac{m}{2\pi}\ln r$	$\frac{m}{2\pi}\theta$
Vortex of circulation Γ	$-\frac{i\Gamma}{2\pi}\operatorname{Log} z$	$\frac{\Gamma}{2\pi}\theta$	$-\frac{\Gamma}{2\pi}\ln r$
Doublet	μ/z	$\mu x/r^2$	$-\mu y/r^2$
Flow past a cylinder $r = a$	$U\left(z + \frac{a^2}{z}\right)$	$U\left(r + \frac{a^2}{r}\right)\cos\theta$	$U\left(r - \frac{a^2}{r}\right)\sin\theta$

Table 2: Standard complex potentials of fluid flow.



solid: streamlines $\psi = 2xy = c$
dashed: equipotentials $\phi = x^2 - y^2 = c$

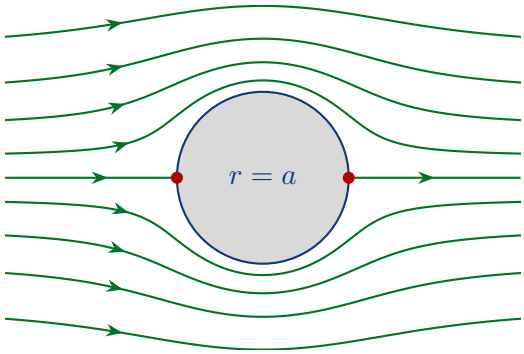


Figure 4: **Left:** flow into a right-angled corner, $\Omega = z^2$. The walls $x = 0$, $y = 0$ are themselves streamlines ($\psi = 0$); streamlines (real part → imaginary part) meet equipotentials at right angles. **Right:** uniform flow past a circular cylinder, $\Omega = U(z + a^2/z)$ with $a = 1$. The circle is the streamline $\psi = 0$; red dots are the stagnation points $z = \pm a$.

Solution. $\phi_{xx} = 6y$, $\phi_{yy} = -6y$, so $\nabla^2\phi = 0$: ϕ is harmonic, hence a possible velocity potential.
$\phi_x = 6xy \Rightarrow \phi_x(z,0) = 0$; $\phi_y = 3x^2 - 3y^2 \Rightarrow \phi_y(z,0) = 3z^2$.
Milne–Thomson: $\Omega'(z) = \phi_x(z,0) - i\phi_y(z,0) = -3iz^2$, so $\Omega(z) = -iz^3 + C$.
Since $z^3 = (x^3 - 3xy^2) + i(3x^2y - y^3)$, we get $-iz^3 = (3x^2y - y^3) - i(x^3 - 3xy^2)$. Hence

$$\boxed{\Omega(z) = -iz^3 + C, \qquad \psi = 3xy^2 - x^3 + c.}$$

Problem 6.2 : *Stream function given (corner flow)*

The stream function of a flow is $\psi = 2xy$. Find the velocity potential, the complex potential and the velocity and speed at the point $(1, 2)$.

Solution. $\psi_y = 2x \Rightarrow \psi_y(z,0) = 2z$; $\psi_x = 2y \Rightarrow \psi_x(z,0) = 0$.
Milne–Thomson (b): $\Omega'(z) = \psi_y(z,0) + i\psi_x(z,0) = 2z$, so $\Omega = z^2 + C$ and $\phi = x^2 - y^2$.
At $z = 1 + 2i$: $\Omega'(z) = 2 + 4i$, so $q_1 + iq_2 = \overline{\Omega'} = 2 - 4i$: velocity $(2, -4)$, speed

$= \sqrt{20} = 2\sqrt{5}$. (Check: $\nabla\phi = (2x, -2y) = (2, -4)$.) The flow is the corner flow of Figure 4 (left).

Problem 6.3 : *Stream function given*

If $\psi = e^{-x}(x\sin y - y\cos y)$ is the stream function of a flow, find the complex potential and the velocity potential.

Solution. $\psi_y = e^{-x}(x\cos y - \cos y + y\sin y) \Rightarrow \psi_y(z,0) = e^{-z}(z-1)$.
$\psi_x = -e^{-x}(x\sin y - y\cos y) + e^{-x}\sin y \Rightarrow \psi_x(z,0) = 0$.
$\Omega'(z) = (z-1)e^{-z}$. Integrating by parts, $\int (z-1)e^{-z}dz = -ze^{-z}$, so

$$\Omega(z) = -z\,e^{-z} + C, \qquad \phi = -\operatorname{Re}(ze^{-z}) = -e^{-x}(x\cos y + y\sin y).$$

Problem 6.4 : *Flow past a cylinder*

Show that $\psi = U\left(y - \dfrac{a^2 y}{x^2+y^2}\right)$ is the stream function of a flow; find its complex potential, the stagnation points and the speed on the cylinder $r = a$.

Solution. With $r^2 = x^2 + y^2$: $\dfrac{\partial}{\partial y}\left(\dfrac{y}{r^2}\right) = \dfrac{x^2-y^2}{r^4}$, $\dfrac{\partial}{\partial x}\left(\dfrac{y}{r^2}\right) = \dfrac{-2xy}{r^4}$.
So $\psi_y(z,0) = U\left(1 - \dfrac{a^2}{z^2}\right)$ and $\psi_x(z,0) = 0$. Hence $\Omega'(z) = U\left(1 - \dfrac{a^2}{z^2}\right)$ and

$$\boxed{\Omega(z) = U\left(z + \frac{a^2}{z}\right)}.$$

On $r = a$: $\psi = U(a-a)\sin\theta = 0$, so the circle is a streamline (a solid cylinder can be placed there).
Stagnation points: $\Omega'(z) = 0 \Rightarrow z = \pm a$.
On the cylinder $z = ae^{i\theta}$: $\Omega' = U(1 - e^{-2i\theta})$, so speed $= |\Omega'| = 2U|\sin\theta|$; maximum $2U$ at the top and bottom.

6.4 Application II : Electrostatic Fields

Definition 6.4 : *Complex electrostatic potential*

In a charge-free region the potential $V(x,y)$ is harmonic. If $\Omega = \phi + i\psi$ is analytic with $\phi = V$, then Ω is the *complex potential*, $\phi = c$ are the *equipotential lines* and $\psi = c$ are the *lines of force* (electric field lines). The field is $E_x + iE_y = -\overline{\Omega'(z)}$ and $|\vec{E}| = |\Omega'(z)|$.

Problem 6.5 : *Equipotentials are circles*

Find the complex potential whose real part is $\phi = \frac{1}{2}\ln(x^2+y^2)$ and describe the lines of force.

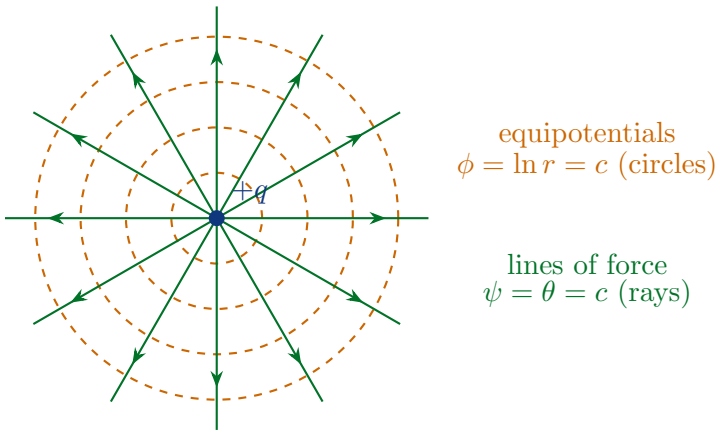


Figure 5: The field of a line charge, $\Omega(z) = -\dfrac{q}{2\pi\varepsilon_0}\operatorname{Log} z$ (shown up to the constant factor). The *real part* $\ln r$ is constant on circles; the *imaginary part* θ is constant on rays – the lines of force leave the charge radially and cut the circles orthogonally. The same picture, read as a fluid flow, is a *source*; read as heat flow, it is a hot wire in a plate.

Solution. $\phi_x = \dfrac{x}{x^2+y^2} \Rightarrow \phi_x(z,0) = \dfrac{1}{z}$; $\quad \phi_y = \dfrac{y}{x^2+y^2} \Rightarrow \phi_y(z,0) = 0$.
$\Omega'(z) = 1/z \Rightarrow \Omega(z) = \operatorname{Log} z + C$. Hence $\psi = \theta = \tan^{-1}(y/x)$: the lines of force are the rays $\theta =$ constant through the origin (Figure 5).

Problem 6.6 : *Dipole potential*

The electrostatic potential in a region is $\phi = \dfrac{x}{x^2+y^2}$. Find the flux function and the complex potential.

Solution. $\phi_x = \dfrac{y^2-x^2}{(x^2+y^2)^2} \Rightarrow \phi_x(z,0) = -\dfrac{1}{z^2}$; $\quad \phi_y = \dfrac{-2xy}{(x^2+y^2)^2} \Rightarrow \phi_y(z,0) = 0$.
$\Omega'(z) = -1/z^2 \Rightarrow \Omega(z) = \dfrac{1}{z} + C$. Since $\dfrac{1}{z} = \dfrac{x-iy}{x^2+y^2}$,

$$\psi = -\frac{y}{x^2+y^2}.$$

The lines of force $\psi = c$ are the circles $x^2 + y^2 + y/c = 0$, all touching the x-axis at the origin.

Problem 6.7 : *Field between hyperbolic electrodes*

The potential is $\phi = x^2 - y^2$. Find the complex potential and the electric field at $(2, 1)$.

Solution. As in Problem 6.2, $\Omega = z^2$ and $\psi = 2xy$. $\Omega'(2+i) = 4+2i$, so $E_x + iE_y = -\overline{\Omega'} = -4+2i$, i.e. $\vec{E} = (-4, 2)$ and $|\vec{E}| = 2\sqrt{5}$. (Check: $-\nabla\phi = (-2x, 2y) = (-4, 2)$.)

6.5 Application III : Analysing AC Circuits

In circuit theory the letter i denotes current, so the imaginary unit is written $j = \sqrt{-1}$.

Phasors – what the real and imaginary parts mean

Definition 6.5 : *Phasor*

A sinusoidal quantity $v(t) = V_m \cos(\omega t + \alpha)$ is written as

$$v(t) = \mathrm{Re}\big[\, V_m e^{j(\omega t+\alpha)}\big] = \mathrm{Re}\big[\, \mathbf{V}\, e^{j\omega t}\big], \qquad \mathbf{V} = V_m e^{j\alpha} = V_m \angle \alpha.$$

The complex constant $\mathbf{V}$ is the *phasor*: its modulus is the amplitude and its argument is the phase.

Physical meaning

- The **real part** of $\mathbf{V}e^{j\omega t}$ is the voltage actually measured on an oscilloscope.
- The **imaginary part** $V_m \sin(\omega t+\alpha)$ is the same signal delayed by a quarter cycle ($90°$); it is carried along only because it turns calculus into algebra:

$$\frac{d}{dt}\big(\mathbf{I}e^{j\omega t}\big) = j\omega\, \mathbf{I}e^{j\omega t}, \qquad \int \mathbf{I}e^{j\omega t}dt = \frac{1}{j\omega}\,\mathbf{I}e^{j\omega t}.$$

- Multiplying a phasor by j rotates it through $+90°$ (a quarter-cycle *lead*).

Complex impedance

Result 6.5 : *Impedance of a series RLC circuit*

For the series circuit of Figure 6, Kirchhoff's voltage law gives

$$L\frac{di}{dt} + Ri + \frac{1}{C}\int i\,dt = V_m \cos \omega t.$$

Try $i = \mathrm{Re}[\mathbf{I}e^{j\omega t}]$. Since the equation is linear with real coefficients, it is enough to solve it for the complex current $\mathbf{I}e^{j\omega t}$ and take the real part:

$$\Big(R + j\omega L + \frac{1}{j\omega C}\Big)\mathbf{I} = V_m \quad \Longrightarrow \quad \boxed{\mathbf{V} = \mathbf{Z}\,\mathbf{I}, \qquad \mathbf{Z} = R + j\Big(\omega L - \frac{1}{\omega C}\Big)}$$

This is **Ohm's law in complex form**; $\mathbf{Z}$ is the *complex impedance*.

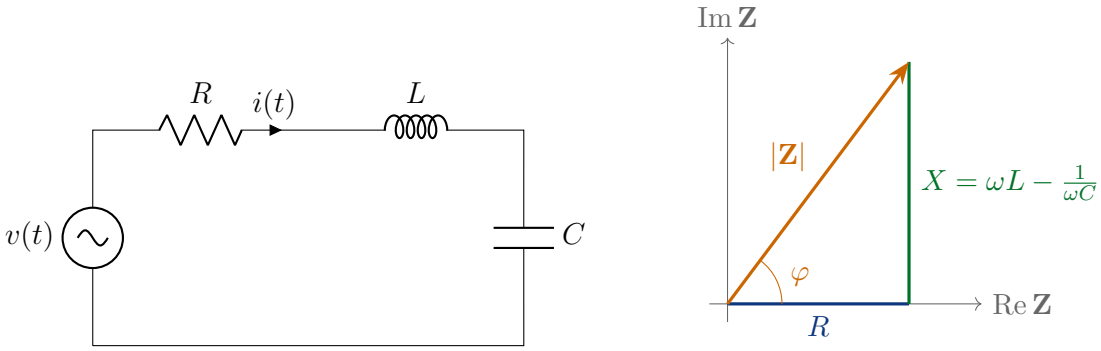


Figure 6: **Left:** series RLC circuit. **Right:** impedance triangle in the complex plane: the real part of $\mathbf{Z}$ is the resistance, the imaginary part is the reactance, and $\varphi = \arg \mathbf{Z}$ is the angle by which the voltage leads the current.

Element	**Law**	**Impedance Z**	**Phase of v relative to i**
Resistor	$v = Ri$	R	in phase (0°)
Inductor	$v = L\,di/dt$	$j\omega L$	v leads by 90°
Capacitor	$i = C\,dv/dt$	$\dfrac{1}{j\omega C} = -\dfrac{j}{\omega C}$	v lags by 90°
Series	–	$\mathbf{Z}_1 + \mathbf{Z}_2 + \cdots$	–
Parallel	–	$\dfrac{1}{\mathbf{Z}} = \dfrac{1}{\mathbf{Z}_1} + \dfrac{1}{\mathbf{Z}_2} + \cdots$	–

Table 3: Impedances of circuit elements. Kirchhoff's laws hold unchanged for phasors.

Meaning of real and imaginary parts in AC circuits

Quantity	**Real part**	**Imaginary part**
$\mathbf{V}e^{j\omega t}$, $\mathbf{I}e^{j\omega t}$	the physical, measured signal	the 90°-shifted companion signal
Impedance $\mathbf{Z} = R + jX$	resistance R: energy *dissipated* as heat	reactance X: energy *stored* and returned ($X > 0$ inductive, $X < 0$ capacitive)
Admittance $\mathbf{Y} = 1/\mathbf{Z} = G + jB$	conductance G	susceptance B
Complex power $\mathbf{S} = P + jQ$	active (average) power P in watts	reactive power Q in VAR

$|\mathbf{Z}| = V_m/I_m$ (ratio of amplitudes); $\arg \mathbf{Z} = \varphi$ (phase difference); $\cos\varphi$ = power factor.

Result 6.6 : *Complex power*

With amplitude phasors $\mathbf{V}, \mathbf{I}$: $\mathbf{S} = \frac{1}{2}\mathbf{V}\,\overline{\mathbf{I}} = P + jQ$, $P = \frac{1}{2}V_m I_m \cos\varphi = \frac{1}{2}I_m^2 R$, $Q = \frac{1}{2}V_m I_m \sin\varphi$.
(With RMS phasors, $\mathbf{S} = \mathbf{V}_{\text{rms}}\overline{\mathbf{I}_{\text{rms}}}$.)

Result 6.7 : *Resonance*

A series RLC circuit is in *resonance* when $\operatorname{Im}\mathbf{Z} = 0$, i.e.

$$\omega_0 L = \frac{1}{\omega_0 C} \quad \Longrightarrow \quad \omega_0 = \frac{1}{\sqrt{LC}}.$$

Then $\mathbf{Z} = R$ (minimum), the current amplitude V_m/R is maximum, v and i are in phase and the power factor is 1. The quality factor is $Q_f = \omega_0 L/R$.

Impedance as an analytic function; the mapping $w = 1/z$

Replacing $j\omega$ by a complex variable s, the impedance $\mathbf{Z}(s) = R + sL + \dfrac{1}{sC}$ and the transfer function $H(s) = \mathbf{V}_{\text{out}}/\mathbf{V}_{\text{in}}$ of any RLC network are rational functions – analytic except at their poles. The steady-state (frequency) response is obtained on the imaginary axis $s = j\omega$; the network is stable when all poles lie in the left half-plane $\operatorname{Re} s < 0$.

Result 6.8 : *Circle diagram (inversion $w = 1/z$)*

If R is fixed and the reactance X varies $(-\infty < X < \infty)$, the impedance $\mathbf{Z} = R + jX$ moves on the vertical line $\operatorname{Re}\mathbf{Z} = R$. The admittance $\mathbf{Y} = 1/\mathbf{Z}$ then moves on the circle

$$\left|\mathbf{Y} - \frac{1}{2R}\right| = \frac{1}{2R},$$

because $w = 1/z$ maps lines not through the origin onto circles through the origin. Hence, for a fixed voltage, the tip of the current phasor $\mathbf{I} = \mathbf{V}\mathbf{Y}$ traces a circle.

Solution. Put $u + jv = \dfrac{1}{R + jX} = \dfrac{R - jX}{R^2 + X^2}$. Then $u^2 + v^2 = \dfrac{1}{R^2 + X^2} = \dfrac{u}{R}$, i.e. $\left(u - \dfrac{1}{2R}\right)^2 + v^2 = \dfrac{1}{4R^2}$. Inductive loads $(X > 0)$ give $v < 0$: the lower semicircle.

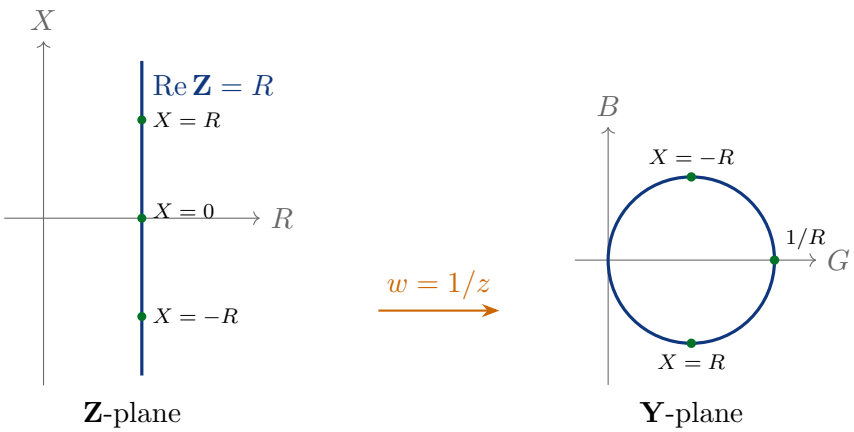


Figure 7: The inversion $w = 1/z$ in circuit analysis: the line of impedances $R + jX$ becomes the admittance circle of diameter $1/R$.

Solved problems on AC circuits

Problem 6.8 : *RL circuit*

A resistor $R = 4\,\Omega$ and an inductor $L = 0.03\,\text{H}$ in series are driven by $v(t) = 10\cos 100t$ volts. Find the steady-state current.

Solution. $\omega L = 3\,\Omega$, so $\mathbf{Z} = 4 + 3j$, $|\mathbf{Z}| = 5$, $\varphi = \tan^{-1}(3/4) = 36.87°$.
$\mathbf{I} = \dfrac{10}{4+3j} = \dfrac{10(4-3j)}{25} = 1.6 - 1.2j = 2\angle -36.87°$.
Taking the real part: $\boxed{i(t) = 2\cos(100t - 36.87°)\ \text{A}}$ – the current lags the voltage (inductive circuit).

Problem 6.9 : *Series RLC circuit and power*

$R = 30\,\Omega$, $L = 0.8\,\text{H}$, $C = 250\,\mu\text{F}$ are in series with $v(t) = 100\cos 100t$ volts. Find the impedance, the current, the power factor and the complex power.

Solution. $\omega L = 80\,\Omega$, $\dfrac{1}{\omega C} = \dfrac{1}{100 \times 250 \times 10^{-6}} = 40\,\Omega$, so $\mathbf{Z} = 30 + j(80-40) = 30 + 40j$, $|\mathbf{Z}| = 50$, $\varphi = 53.13°$.
$\mathbf{I} = \dfrac{100}{30+40j} = \dfrac{100(30-40j)}{2500} = 1.2 - 1.6j = 2\angle -53.13°$, so $i(t) = 2\cos(100t - 53.13°)$ A.
Power factor $= \cos\varphi = 30/50 = 0.6$ (lagging).
$\mathbf{S} = \frac{1}{2}\mathbf{V}\bar{\mathbf{I}} = \frac{1}{2}(100)(1.2 + 1.6j) = 60 + 80j$: active power $P = 60$ W, reactive power $Q = 80$ VAR. (Check: $\frac{1}{2}I_m^2 R = \frac{1}{2}(4)(30) = 60$ W.)

Problem 6.10 : *Parallel circuit*

A resistor $R = 10\,\Omega$ is in parallel with a capacitor of reactance $10\,\Omega$ at the operating frequency. For a voltage phasor $\mathbf{V} = 100\angle 0°$, find the equivalent impedance and the total current.

Solution. $\mathbf{Y} = \dfrac{1}{R} + j\omega C = 0.1 + 0.1j$, so $\mathbf{Z} = \dfrac{1}{\mathbf{Y}} = \dfrac{1}{0.1(1+j)} = 5 - 5j\ \Omega$.
$\mathbf{I} = \mathbf{V}\mathbf{Y} = 100(0.1 + 0.1j) = 10 + 10j = 14.14\angle 45°$ A: the current *leads* the voltage by 45° (capacitive circuit).

Problem 6.11 : *Resonance*

For the circuit of Problem 6.9, find the resonant frequency, the current amplitude at resonance and the quality factor.

Solution. $LC = 0.8 \times 250 \times 10^{-6} = 2 \times 10^{-4}$, so $\omega_0 = 1/\sqrt{LC} = 70.71$ rad/s ($f_0 = \omega_0/2\pi = 11.25$ Hz).
At resonance $\mathbf{Z} = R = 30\,\Omega$, so $I_m = 100/30 = 3.33$ A (maximum), in phase with v.
$Q_f = \omega_0 L/R = 70.71 \times 0.8/30 = 1.89$.

Problem 6.12 : *Frequency response of an RC filter*

For an RC low-pass filter (R in series, output across C), show that $H(s) = \dfrac{1}{1+sRC}$ and find the gain and phase at $\omega = 1000$ rad/s when $R = 1\,\text{k}\Omega$, $C = 1\,\mu\text{F}$.

Solution. Voltage divider: $H = \dfrac{1/(sC)}{R + 1/(sC)} = \dfrac{1}{1+sRC}$ – analytic everywhere except the pole $s = -1/RC$ (left half-plane, stable). With $RC = 10^{-3}$ and $s = j\omega = 1000j$: $H = \dfrac{1}{1+j} = \dfrac{1-j}{2}$.
Gain $|H| = 1/\sqrt{2} = 0.707$ (-3 dB), phase $\arg H = -45°$ (output lags input).

Problem 6.13 : *Locus of current (circle diagram)*

A series circuit has fixed $R = 5\,\Omega$ and variable reactance X, with $\mathbf{V} = 10\angle 0°$. Show that the tip of the current phasor moves on a circle and find its centre, radius and the maximum current.

Solution. $\mathbf{I} = \dfrac{10}{5+jX} = 10\,\mathbf{Y}$. By Result 6.8, $\mathbf{Y}$ lies on $|\mathbf{Y} - \frac{1}{10}| = \frac{1}{10}$, hence $\boxed{|\mathbf{I} - 1| = 1}$: centre $1 + 0j$, radius 1. Maximum $|\mathbf{I}| = 2$ A (the diameter), reached when $X = 0$ (resonance).

6.6 Application IV : Steady-State Heat Transfer

Concepts

Definition 6.6 : *Complex temperature*

By *Fourier's law* the heat flux in a body of conductivity k is $\vec{q} = -k\nabla T$. In the steady state with no heat sources, $\nabla \cdot \vec{q} = 0$, so the temperature satisfies

$$\nabla^2 T = T_{xx} + T_{yy} = 0$$

in a thin plate with insulated faces (or a long body with no variation along its length). If $F(z) = T(x,y) + i\,\psi(x,y)$ is analytic, F is the *complex temperature* (complex heat potential):

- $T = \operatorname{Re} F$ is the **temperature**; the curves $T = c$ are **isotherms**;
- $\psi = \operatorname{Im} F$ is the **heat-flow function**; the curves $\psi = c$ are **heat-flow lines**, along which heat travels.

Result 6.9 : *Heat flux from the complex temperature*

$$q_x + i\,q_y = -k\,\overline{F'(z)}, \qquad |\vec{q}| = k\,|F'(z)|.$$

Heat always flows from hot to cold, perpendicular to the isotherms. The heat crossing any curve from A to B (per unit thickness, per unit time) is $k\,|\psi(B) - \psi(A)|$.

Result 6.10 : *Conformal invariance of Laplace's equation*

If $\Phi(u, v)$ is harmonic in a domain D' of the w-plane and $w = f(z) = u + iv$ is analytic and maps D onto D', then $\Phi\big(u(x, y), v(x, y)\big)$ is harmonic in D. Boundary values are carried over point by point.

Strategy for heat-conduction (Dirichlet) problems

1. Choose a conformal map $w = f(z)$ (e^z, $\sin z$, $1/z$, a Möbius map, z^n, $\operatorname{Log} z$) that sends the given region to a simple one: a half-plane, a strip or an annulus.
2. Solve there, where the answer is elementary: strip: $T = a + bu$; half-plane with a jump on the boundary: $T = a + b \operatorname{Arg} w$; annulus: $T = a + b \ln |w|$.
3. Substitute $w = f(z)$ back. The real part of the resulting analytic function is T; its imaginary part gives the heat-flow lines.

Solved problems on heat transfer

Problem 6.14 : *Temperature given – find heat flow*

The temperature in a thin plate is $T = x^3 - 3xy^2 + 3x^2 - 3y^2 + 1$. Verify that it is a possible steady temperature, find the complex temperature and the heat-flow function, and find the heat flux at $(1, 1)$.

Solution. $T_{xx} + T_{yy} = (6x + 6) + (-6x - 6) = 0$ ✓.
$T_x = 3x^2 - 3y^2 + 6x \Rightarrow T_x(z, 0) = 3z^2 + 6z$; $T_y = -6xy - 6y \Rightarrow T_y(z, 0) = 0$.
$F'(z) = 3z^2 + 6z \Rightarrow F(z) = z^3 + 3z^2 + 1 + iC$ (the constant 1 matches T). Hence

$$\psi = \operatorname{Im}(z^3 + 3z^2) = 3x^2y - y^3 + 6xy \quad \text{(heat-flow lines } \psi = c\text{)}.$$

At $z = 1 + i$: $F'(1 + i) = 3(2i) + 6(1 + i) = 6 + 12i$, so $q_x + iq_y = -k(6 - 12i)$: $\vec{q} = (-6k,\ 12k)$, $|\vec{q}| = 6\sqrt{5}\,k$.

Problem 6.15 : *Heat-flow function given – find temperature*

The heat-flow function in a plate is $\psi = 2x - 2xy$. Find the complex temperature and the temperature distribution.

Solution. $\psi_{xx} + \psi_{yy} = 0$ ✓. $\psi_y = -2x \Rightarrow \psi_y(z, 0) = -2z$; $\psi_x = 2 - 2y \Rightarrow \psi_x(z, 0) = 2$.
Milne–Thomson (b): $F'(z) = -2z + 2i \Rightarrow F(z) = -z^2 + 2iz + C$.
$-z^2 + 2iz = (y^2 - x^2 - 2y) + i(2x - 2xy)$, so $\boxed{T = y^2 - x^2 - 2y + C}$.

Problem 6.16 : *Half-plane with two boundary temperatures* ($\operatorname{Log} z$)

Find the steady temperature in the upper half-plane $y > 0$ if the positive x-axis is kept at T_1 and the negative x-axis at T_2.

Solution. $\operatorname{Log} z = \ln r + i\theta$ is analytic in $y > 0$, so $\theta = \operatorname{Im} \operatorname{Log} z$ is harmonic with $\theta = 0$ on $x > 0$ and $\theta = \pi$ on $x < 0$. Take $T = A + B\theta$: the boundary conditions give $A = T_1$, $B = (T_2 - T_1)/\pi$:

$$T = T_1 + \frac{T_2 - T_1}{\pi}\,\theta = T_1 + \frac{T_2 - T_1}{\pi}\tan^{-1}\frac{y}{x}, \qquad F(z) = T_1 - \frac{i(T_2 - T_1)}{\pi}\operatorname{Log} z.$$

Isotherms: rays $\theta = c$. Heat-flow lines: $\psi = -\frac{T_2 - T_1}{\pi}\ln r = c$, i.e. semicircles $r = c$ (Figure 8, left).

Problem 6.17 : *Quarter-plane (90° wedge)*

The region $x > 0,\ y > 0$ has the edge $y = 0$ at 0°C and the edge $x = 0$ at 100°C. Find the temperature.

Solution. $w = z^2$ maps the quadrant onto the upper half-plane, and $\operatorname{Arg} w = 2\theta$. By Problem 6.16 (with $T_1 = 0$, $T_2 = 100$), $T = \frac{100}{\pi}\operatorname{Arg} w = \frac{200}{\pi}\,\theta = \frac{200}{\pi}\tan^{-1}\frac{y}{x}$. Check: $\theta = 0 \Rightarrow T = 0$; $\theta = \pi/2 \Rightarrow T = 100$ ✓.

Problem 6.18 : *Insulated pipe (annulus)*

A long pipe has inner radius $a = 1$ and outer radius $b = 2$ (cm). The inner surface is at 100°C and the outer at 20°C. Find the temperature in the wall and the heat loss per unit length.

Solution. By symmetry T depends only on r; $T = A\ln r + B = \operatorname{Re}(A\operatorname{Log} z + B)$ is harmonic. $r = 1$: $B = 100$; $r = 2$: $A\ln 2 + 100 = 20 \Rightarrow A = -80/\ln 2$. Therefore

$$T = 100 - \frac{80\ln r}{\ln 2}, \qquad F(z) = 100 - \frac{80}{\ln 2}\operatorname{Log} z.$$

Isotherms: circles $r = c$; heat-flow lines: rays $\theta = c$ (Figure 8, centre). Radial flux $q_r = -k\,dT/dr = \frac{80k}{r\ln 2}$; heat loss per unit length $Q = 2\pi r\,q_r = \frac{160\pi k}{\ln 2} \approx 725.2\,k$ (independent of r, as it must be).
In general $Q = \dfrac{2\pi k\,(T_1 - T_2)}{\ln(b/a)}$.

Problem 6.19 : *Strip (e^z)*

Find the temperature in the strip $0 < y < \pi$ whose edges $y = 0$ and $y = \pi$ are held at 0° and 100° respectively, using the mapping $w = e^z$.

Solution. $w = e^z = e^x e^{iy}$ maps the strip onto the upper half-plane with $\operatorname{Arg} w = y$; $y = 0 \mapsto$ positive u-axis and $y = \pi \mapsto$ negative u-axis. From Problem 6.16, $T = \frac{100}{\pi}\operatorname{Arg} w = \frac{100\,y}{\pi}$: the temperature rises linearly across the strip; isotherms are the lines $y = c$, heat flows along the lines $x = c$.

Problem 6.20 : *Region between two tangent circles ($1/z$)*

The circles $|z-1|=1$ and $|z-2|=2$ touch at the origin. The inner circle is kept at $0°$ and the outer one at T_0. Find the temperature in the crescent-shaped region between them.

Solution. Under $w=1/z$, $u=\dfrac{x}{x^2+y^2}$. The circle $x^2+y^2=2x$ maps to the line $u=\frac{1}{2}$, and $x^2+y^2=4x$ to $u=\frac{1}{4}$: the crescent becomes the strip $\frac{1}{4}<u<\frac{1}{2}$. There T is linear in u: $T=T_0(2-4u)$ ($u=\frac{1}{2}\Rightarrow 0$, $u=\frac{1}{4}\Rightarrow T_0$). Returning to z:

$$T=T_0\left(2-\frac{4x}{x^2+y^2}\right), \qquad F(z)=T_0\left(2-\frac{4}{z}\right).$$

The isotherms are the circles $x^2+y^2=cx$, all tangent to the y-axis at the origin.

Problem 6.21 : *Semi-infinite strip ($\sin z$)*

Find the steady temperature in the semi-infinite strip $-\frac{\pi}{2}<x<\frac{\pi}{2}$, $y>0$, when the vertical sides are kept at $0°$ and the base $y=0$ at $1°$ (unit temperature).

Solution. *Step 1 (map).* $w=\sin z$ maps the strip onto the upper half w-plane: the base $-\frac{\pi}{2}<x<\frac{\pi}{2}$ goes to the segment $-1<u<1$, and the sides $x=\pm\frac{\pi}{2}$ go to $u\geq 1$ and $u\leq -1$.
Step 2 (half-plane). In the w-plane we need $T=1$ on $|u|<1$ and $T=0$ on $|u|>1$. The harmonic function

$$T=\frac{1}{\pi}\big[\operatorname{Arg}(w-1)-\operatorname{Arg}(w+1)\big]$$

does this: on $|u|<1$, $\operatorname{Arg}(w-1)=\pi$ and $\operatorname{Arg}(w+1)=0$, giving 1; on $u>1$ both are 0; on $u<-1$ both are π. Geometrically, πT is the angle subtended at w by the hot segment.
Step 3 (back to z). Using $\sin z=\sin x\cosh y+i\cos x\sinh y$ and simplifying $\tan\big(\operatorname{Arg}(w-1)-\operatorname{Arg}(w+1)\big)$,

$$\boxed{T(x,y)=\frac{2}{\pi}\tan^{-1}\left(\frac{\cos x}{\sinh y}\right)}$$

Check: $x=\pm\frac{\pi}{2}\Rightarrow T=0$; $y\to 0^+\Rightarrow T\to\frac{2}{\pi}\cdot\frac{\pi}{2}=1$; $y\to\infty\Rightarrow T\to 0$ ✓. The isotherm $T=c$ is the curve $\sinh y=\cos x\,/\tan(\pi c/2)$ (Figure 8, right).

Note

The *same* mathematical solution serves every field in Table 1. For example, the pipe solution of Problem 6.18 is also the potential between two coaxial cylindrical conductors, and Problem 6.16 is also the potential of two semi-infinite plates at voltages T_1, T_2 separated by a small gap at the origin.

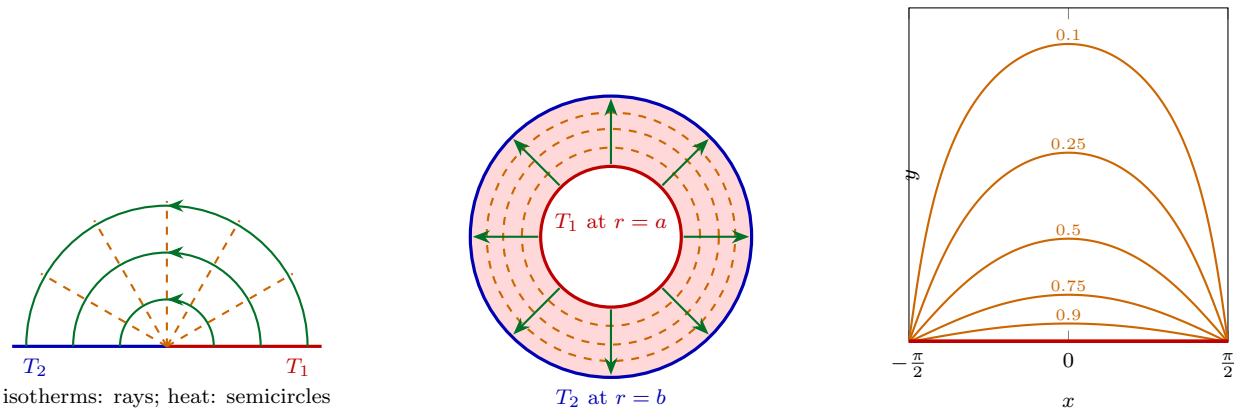


Figure 8: Heat-transfer fields. **Left:** half-plane with boundary temperatures T_1, T_2 (Problem 6.16): isotherms are rays (real part of F), heat flows along semicircles (imaginary part of F). **Centre:** pipe wall (Problem 6.18): circular isotherms, radial heat flow. **Right:** semi-infinite strip heated at the base (Problem 6.21): isotherms $T = \frac{2}{\pi}\tan^{-1}(\cos x/\sinh y) = c$.

6.7 Summary

Real and imaginary parts at a glance

- **Potential fields** (fluid, electrostatic, heat): $\operatorname{Re} f$ = potential / temperature, $\operatorname{Im} f$ = stream / flux / heat-flow function; their level curves are orthogonal; $\overline{f'(z)}$ is the gradient vector of $\operatorname{Re} f$.
- **AC circuits**: Re of a phasor signal = the measured signal; $\operatorname{Re}\mathbf{Z}$ = resistance (dissipation), $\operatorname{Im}\mathbf{Z}$ = reactance (storage); $\operatorname{Re}\mathbf{S}$ = active power, $\operatorname{Im}\mathbf{S}$ = reactive power; $\arg\mathbf{Z}$ = phase.
- **Construction**: from a given ϕ or ψ, use Milne–Thomson to get the complex potential, then read off the other part.
- **Mappings**: $\operatorname{Log} z$ (wedges, annuli), e^z (strips → half-plane), $\sin z$ (semi-infinite strip → half-plane), $1/z$ and Möbius maps (circles ↔ lines; admittance circle diagram).

6.8 Practice Exercises (with answers)

1. The velocity potential of a flow is $\phi = x^2 - y^2 + x$. Find the stream function.
   Ans. $\Omega = z^2 + z$, $\psi = 2xy + y$.

2. Show that $\phi = e^x \cos y$ is a possible electrostatic potential and find the flux function.
   Ans. $\Omega = e^z$, $\psi = e^x \sin y$.

3. The temperature in a plate is $T = y^3 - 3x^2y$. Find the heat-flow function.
   Ans. $F = iz^3$, $\psi = x^3 - 3xy^2$.

4. A series circuit has $R = 6\,\Omega$, $X_L = 10\,\Omega$, $X_C = 2\,\Omega$ and $V_m = 100$ V. Find I_m, the phase angle and the average power.
   Ans. $\mathbf{Z} = 6 + 8j$, $I_m = 10$ A, $\varphi = 53.13°$ lag, $P = 300$ W.

5. Find the resonant angular frequency when $L = 0.5$ H and $C = 20\,\mu$F.
   Ans. $\omega_0 = 316.2$ rad/s.

6. A pipe has $a = 1$, $b = e$, inner temperature $50°$ and outer temperature $0°$. Find T and the heat loss per unit length.
   Ans. $T = 50(1 - \ln r)$, $Q = 100\pi k$.

7. In the quadrant $x > 0, y > 0$ the edge $y = 0$ is at $0°$ and $x = 0$ at $50°$. Find T.
   Ans. $T = \dfrac{100}{\pi} \tan^{-1} \dfrac{y}{x}$.

8. Show that under $w = 1/z$ the line $x = 2$ maps to the circle $u^2 + v^2 = \frac{u}{2}$ and interpret it as an admittance locus.